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Questions

Q1.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The risk of developing cardiovascular disease (CVD) can be increased by a number of factors including diet and a history of thrombosis.

Thrombosis is the development of a blood clot in a blood vessel. Blood clots can develop in veins and then move through the circulatory system into the coronary artery.

(a) A diet high in saturated triglycerides raises the levels of cholesterol in the blood.

(i) What is the ratio of glycerol to fatty acid in a triglyceride molecule?

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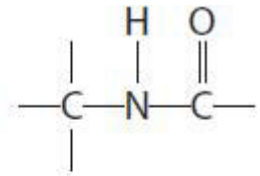
(1)

- ☐ **A** 1:1
- ☐ **B** 1:3
- ☐ **C** 3:1
- ☐ **D** 3:3

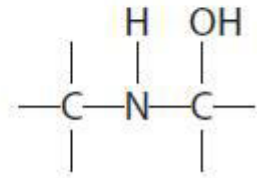
(ii) Which diagram shows the bond that joins a glycerol molecule to a fatty acid in a triglyceride?

(1)

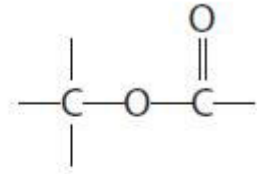
☐ **A**



☐ **B**

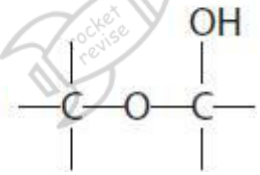


☐ **C**



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☐ **D**



(iii) The table gives some information about four fatty acids.

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Fatty acid	Number of double bonds between carbon atoms	Number of carbon atoms
butyric	0	4
stearic	0	18
palmitoleic	1	16
linoleic	2	18

Explain which of these fatty acids would have the lowest risk of causing CVD, if included in a diet in equal masses.

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(b) Anticoagulants, antiplatelets and thrombolytics are drugs used to treat blood clots.

(i) One anticoagulant binds to the active site of thrombin.
Explain how this drug reduces blood clotting.

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(ii) Molecules on the surface of platelets enable them to bind to other molecules.
One of the antiplatelet drugs affects molecules on the surface of platelets.
Explain how this drug reduces blood clotting.

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(iii) One thrombolytic drug converts plasminogen into the active enzyme, plasmin. Plasmin breaks down fibrin.

Explain how this drug reduces the formation of blood clots.

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(Total for question = 10 marks)

Q2.

Dietary antioxidants may reduce the risk of cardiovascular disease (CVD).

(a) Explain how dietary antioxidants reduce the risk of CVD.

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(b) Chocolate contains high concentrations of a group of antioxidants called flavonoids.

It has been suggested that eating chocolate could reduce the risk of CVD.
The table shows some information about two types of chocolate.

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Type of chocolate	Mass of flavonoids / mg per 100 g of chocolate	Energy content / kJ per 100 g of chocolate
milk chocolate	70	2345
dark chocolate	170	1800

(i) Explain whether eating dark chocolate is likely to reduce the risk of CVD more than eating milk chocolate. Use the information in the table to support your answer.

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(ii) The table gives some information about ingredients in three different brands of milk chocolate and dark chocolate.

Ingredient	Percentage of ingredient in milk chocolate (%)			Percentage of ingredient in dark chocolate (%)		
	brand 1	brand 2	brand 3	brand 1	brand 2	brand 3
cocoa mass	16	16	16	40	40	40
sugar	40	40	40	50	50	50
milk fat	0	3	0	0	3	1
whole milk powder	20	20	20	0	0	0

Use the information in both tables to identify which ingredient contains the most flavonoids.

(1)

- ☒ **A** cocoa mass
- ☒ **B** milk fat
- ☒ **C** sugar
- ☒ **D** whole milk powder

(iii) Describe how a study could be designed to collect valid and reliable data on

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the effects of eating chocolate on the risk of CVD.

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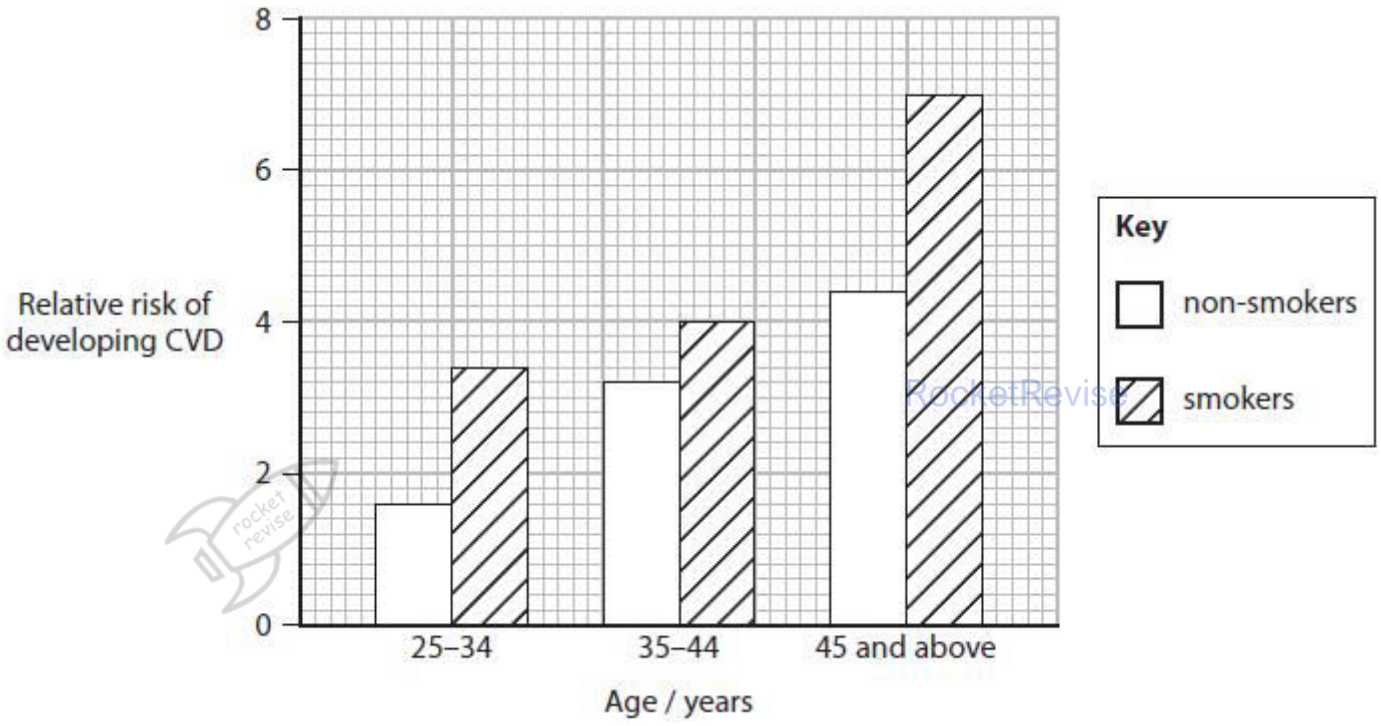
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(Total for question = 11 marks)

Q3.

The graph shows the relative risk of developing cardiovascular disease (CVD) for non-smokers and for smokers.



(a) (i) Describe the conclusions that can be made about the risk factors for CVD.

Use the information in the graph to support your answer.

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(ii) The information shown in this graph was collected more than 25 years ago.
Explain how this graph might appear if it showed data collected last year.

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(b) The number of people in the world who die from CVD each year is 17.7 million.

This is 31% of all total deaths.
How many deaths are there in the world each year?

(1)

- ☒ **A** 0.571×10^6
- ☒ **B** 5.71×10^6
- ☒ **C** 57.1×10^6
- ☒ **D** 571.00×10^6

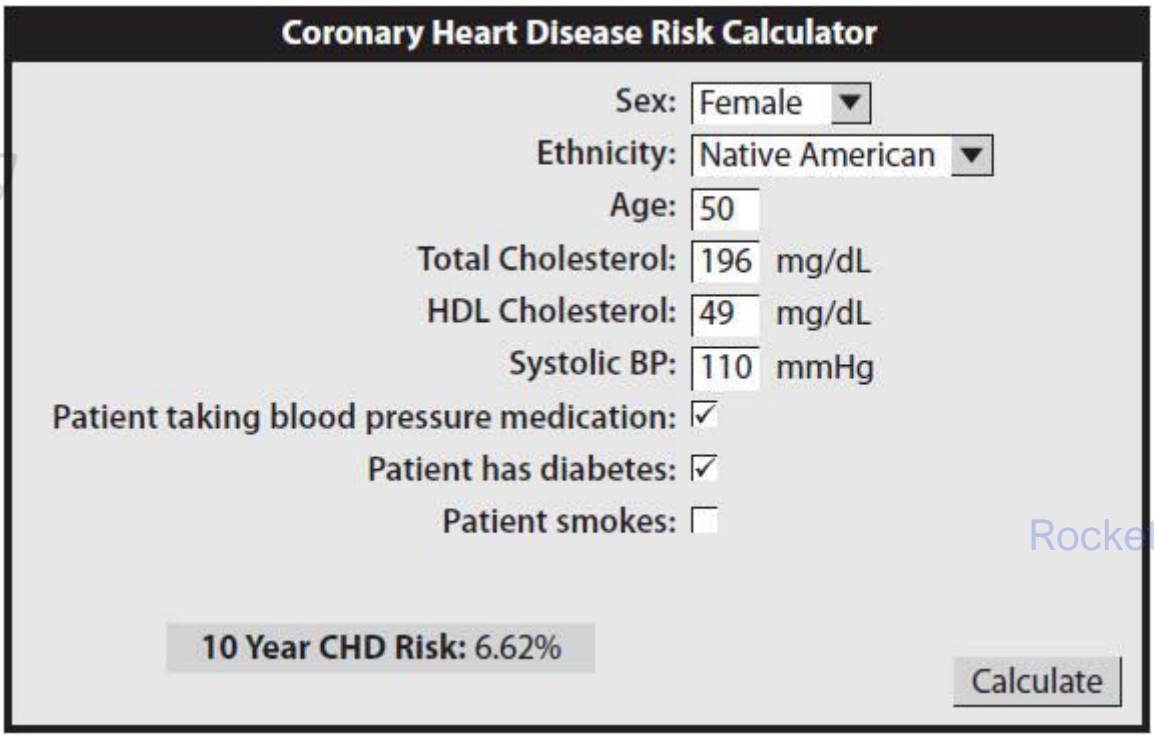
(Total for question = 5 marks)

Q4.

The diagram shows a coronary heart disease (CHD) risk calculator.

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(a) Which of the following is a medication to control blood pressure?

(1)

- ☒ **A** anticoagulant
- ☒ **B** antihypertensive
- ☒ **C** platelet inhibitor
- ☒ **D** statin

(b) Explain why the risk calculator takes into account total cholesterol and HDL cholesterol levels.

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(c) Explain why the 10 Year CHD Risk would change if this person ticked the box next to the 'Patient smokes' on the risk calculator.

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(d) (i) State one other factor that the risk calculator should include to improve the accuracy of the value for the 10 Year CHD Risk.

(1)

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(ii) Predict the effect that taking this factor into account would have on the value for the 10 Year CHD Risk.

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(iii) A person used this risk calculator.
Explain why the value obtained for the 10 Year CHD Risk may be an underestimate.

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(Total for question = 10 marks)

Q5.

Obesity increases the risk of cardiovascular disease (CVD).

One way to reduce obesity is to lose weight by changing eating habits.

(a) Name **two** obesity indicators used by scientists.

(1)

1 .....

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(b) Glucomannan is a dietary supplement claimed to aid weight loss.

Glucomannan is a branched polysaccharide similar in structure to amylopectin.

(i) Which glycosidic bonds are responsible for the branching in glucomannan?

(1)

- ☒ **A** 1-4 only
- ☒ **B** 1-6 only
- ☒ **C** both 1-4 and 1-6
- ☒ **D** neither 1-4 nor 1-6

(ii) In the presence of water, glucomannan swells to form a semi-solid gel.

The diagram shows a stomach with glucomannan present and a stomach without glucomannan.



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Suggest how glucomannan aids weight loss.

(1)

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(iii) If glucomannan could be digested it would cause a gain in weight.

Explain why glucomannan could cause a gain in weight.

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(c) A study compared weight loss in two groups of women.

One group of women had a low-fat diet and the other group of women had a very low-carbohydrate diet.

(i) The table shows the results of this study.

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Number of weeks on the diet	Mean body mass of the group of women / kg	
	Group on the low-fat diet	Group on the very low-carbohydrate diet
0	92.5	91.1
2	90.6	88.3
4	89.9	87.5
6	89.0	85.6
8	88.8	84.5
10	88.3	84.1
12	88.2	83.0

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Many studies claim that a low-carbohydrate diet can result in two to three times as much weight loss as a low-fat diet.
Determine the extent to which this study supports this claim.

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(ii) Very low-carbohydrate diets may increase cardiovascular risk factors.
Give **two** factors, other than weight loss, that should have been monitored in this study.

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(Total for question = 10 marks)

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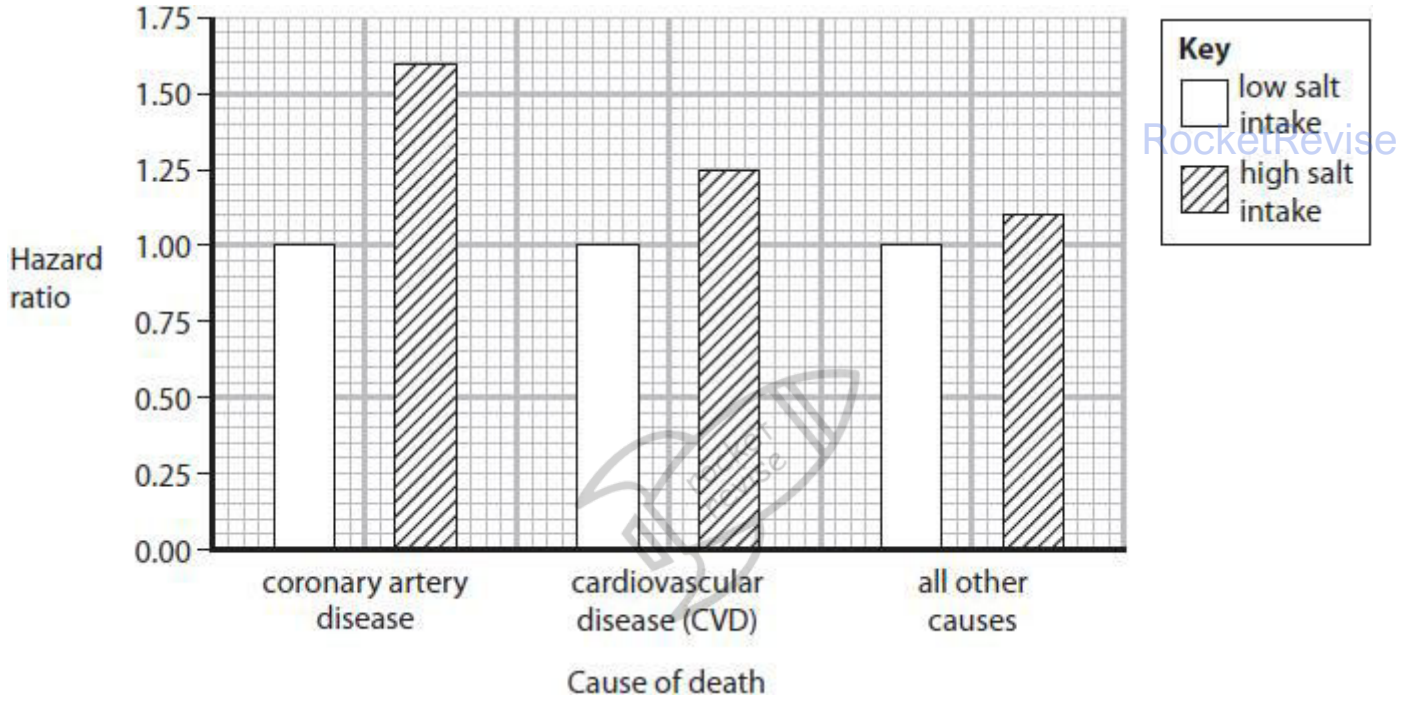
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Q6.

A diet high in salt has been shown to increase the risk of some diseases.

The graph shows the results of one study into the increased risk of death related to a diet high in salt.

An increased risk of death is associated with a high hazard ratio.



(a) State **two** conclusions that can be drawn from these results.

(2)

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(b) The values for the hazard ratio were adjusted for a number of factors, including age and smoking.

(i) Explain why age and smoking were taken into account when determining the hazard ratio.

(3)

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(ii) State **two** factors, other than age and smoking, that may have been considered when adjusting the values for the hazard ratio.

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(c) Suggest why the hazard ratio for low salt intake was 1.00 for each cause of death.

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(d) (i) The results of this study suggest that there is a correlation between salt intake and cause of death.

State the meaning of the term **correlation**.

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(ii) Suggest why studies of this type are unreliable.

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(Total for question = 12 marks)

Q7.

Obesity increases the risk of cardiovascular disease (CVD).

Body mass index (BMI), waist-to-hip ratio (WHR) and skinfold thickness are indicators of obesity.

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(a) The table shows some measurements taken from two females, female **J** and female **K**.

Female	Height / cm	Mass / kg	Waist / cm	Hips / cm	BMI	WHR
J	155	59.1	80	100	25	0.80
K	155			125	36	0.80

(i) Calculate the waist size of female **K**.

(1)

Answer ..... cm

(ii) The formula for calculating BMI is:

$$\text{BMI} = \frac{\text{mass in kg}}{(\text{height in m})^2}$$

Calculate the mass of female **K**, using the data in the table.

(3)

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Answer ..... kg

(iii) Comment on the risk of developing CVD in these two women.

(2)

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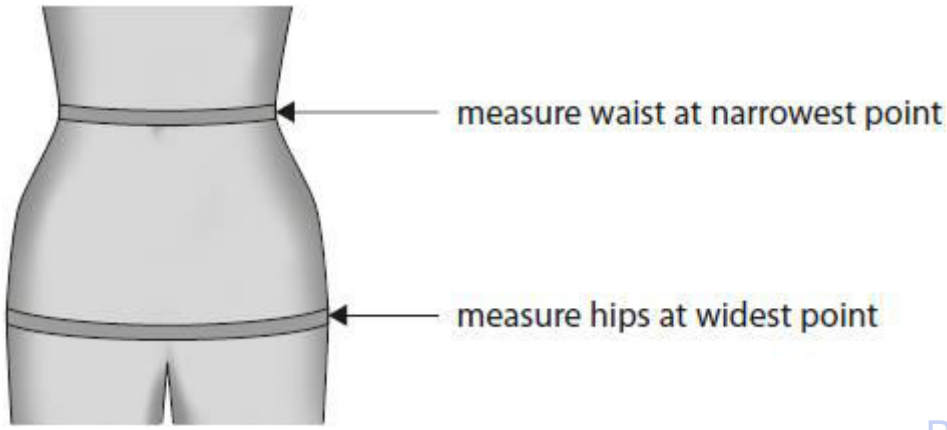
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(iv) The diagram shows how the waist and hip measurements should be taken, using a tape measure.



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Explain how the way a person takes these measurements could produce an underestimate of their risk of CVD.

(2)

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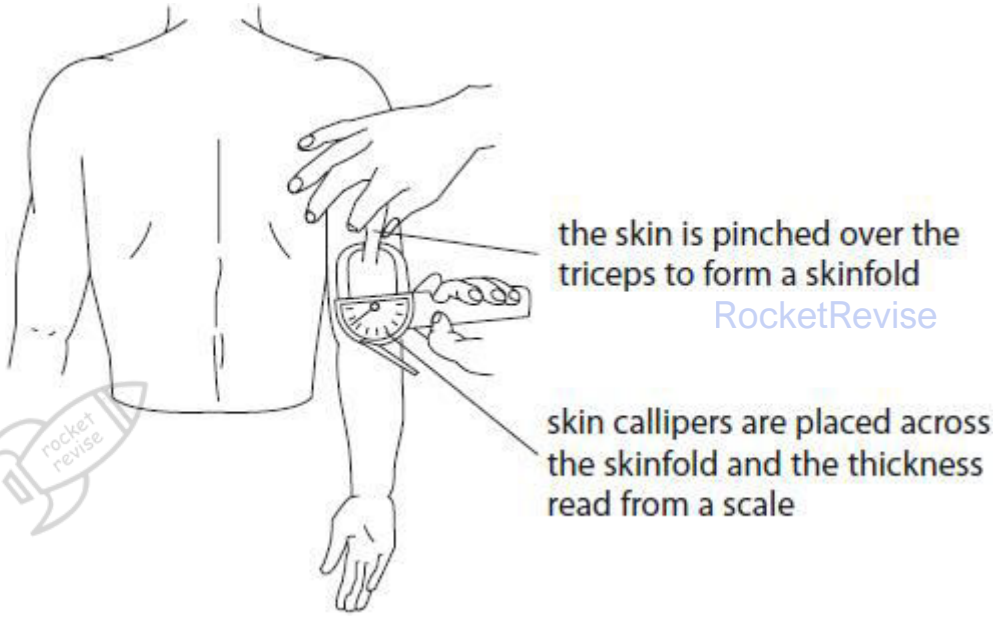
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(b) The diagram shows how a skinfold thickness measurement is taken over the triceps muscle at the back of the arm.



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Skinfold thickness measures the thickness of the layer of fat under the skin.
Measurements are taken from several sites on the body.
The table shows the skinfold thickness at four sites on a 42-year-old female.

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Site	Skinfold thickness / a.u.
over the biceps at the front of the arm	0.63
over the triceps at the back of the arm	0.82
under the shoulder blade at the back of the neck	0.65
above the hip bone at the side of the body	0.82

(i) Suggest **two** reasons why the skinfold thickness values are different at each site on the body. Assume that the skin callipers have been used correctly.

(2)

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(ii) The table shows a body fat interpretation chart.
The values in the table are the means of the four skinfold thickness measurements.

Age	Level of body fat			
	Low	Moderate	High	Very high
20 to 29	<0.71	0.71 to 0.77	0.78 to 0.82	>0.82
30 to 39	<0.72	0.72 to 0.78	0.79 to 0.84	>0.84
40 to 49	<0.73	0.73 to 0.79	0.80 to 0.87	>0.87
50 to 59	<0.74	0.74 to 0.81	0.82 to 0.88	>0.88

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Determine the level of body fat of this female.

(1)

Answer .....

(Total for question = 11 marks)

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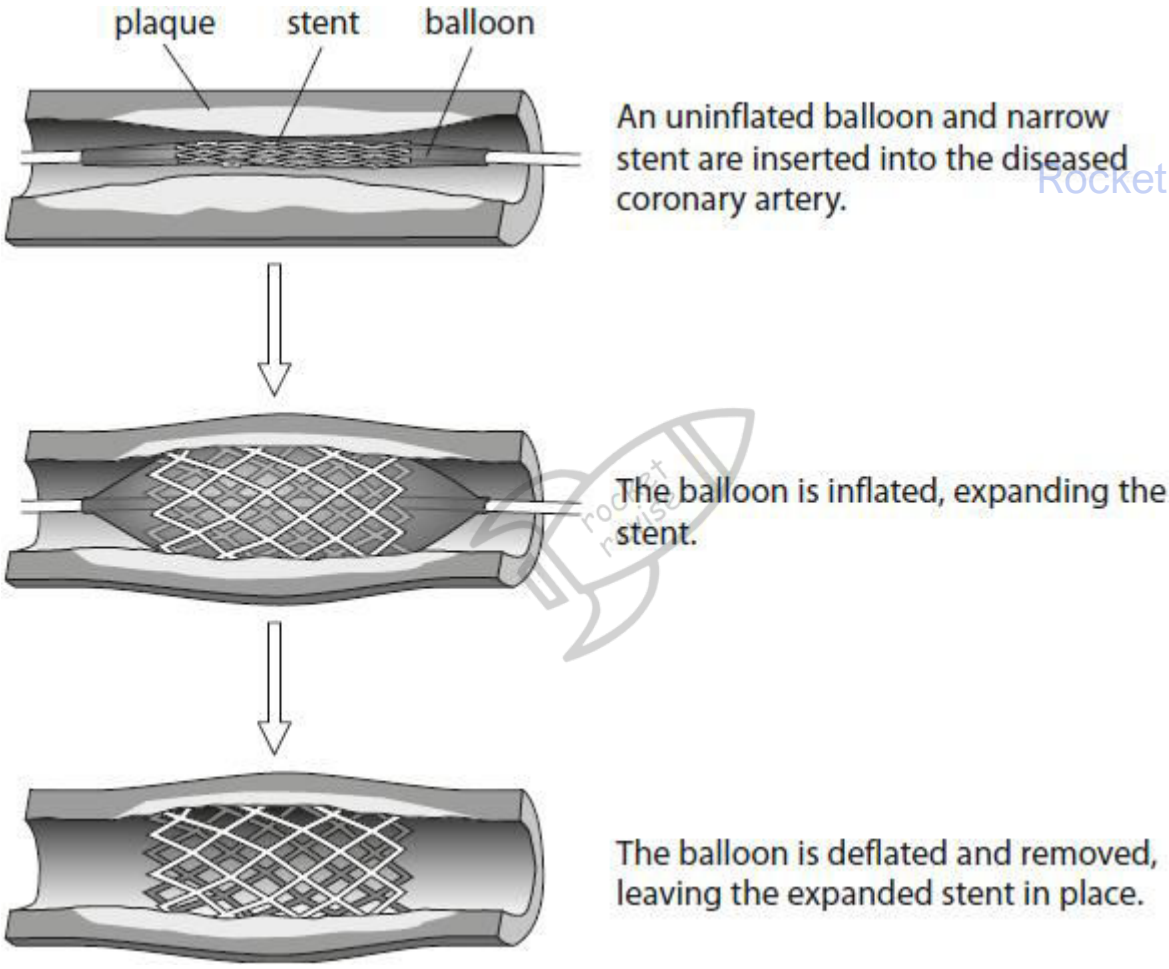
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Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Stents are used in the treatment of atherosclerosis.

(a) The diagram shows how stents are positioned in a diseased coronary artery.



Explain why a stent is used in the treatment of atherosclerosis in a coronary artery.

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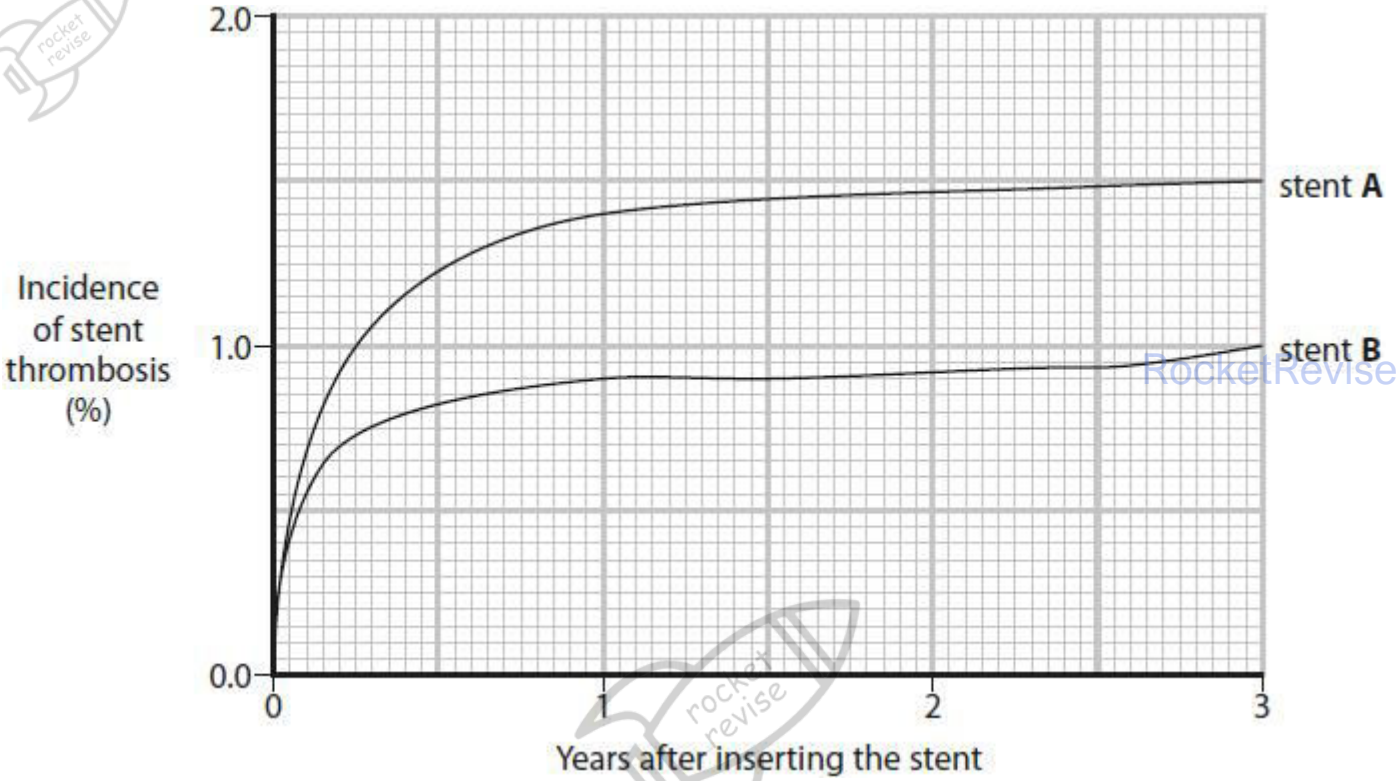
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(b) Inserting a stent can damage the artery.

This damage can result in stent thrombosis if the blood clotting process is stimulated.
A study looked at the damage caused by two different types of stent, stent **A** and stent **B**.
The graph shows the incidence of stent thrombosis found in this study.



(i) In this study, 800 patients had stent **A** inserted and 400 patients had stent **B** inserted.
Calculate the difference in the number of patients who developed stent thrombosis three years after inserting the stent.

(2)

Answer .....

(ii) State how the formation of a blood clot in the coronary artery could cause the death of the patient.

(1)

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(iii) Which components of the blood clotting process are enzymes in their active form?

(1)

- ☒ **A** prothrombin and thromboplastin only
- ☒ **B** prothrombin and thrombin only
- ☒ **C** prothrombin, thrombin and thromboplastin
- ☒ **D** thrombin and thromboplastin only

(iv) Which components of the blood clotting process are soluble in blood plasma?

(1)

- ☒ **A** fibrin and fibrinogen
- ☒ **B** fibrin and thromboplastin

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☒ **C** fibrinogen and thromboplastin
☒ **D** fibrinogen only

(v) Stent **B** contains a drug to prevent stent thrombosis.
Suggest one type of drug that could be used in stent **B**.

(1)

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(Total for question = 9 marks)

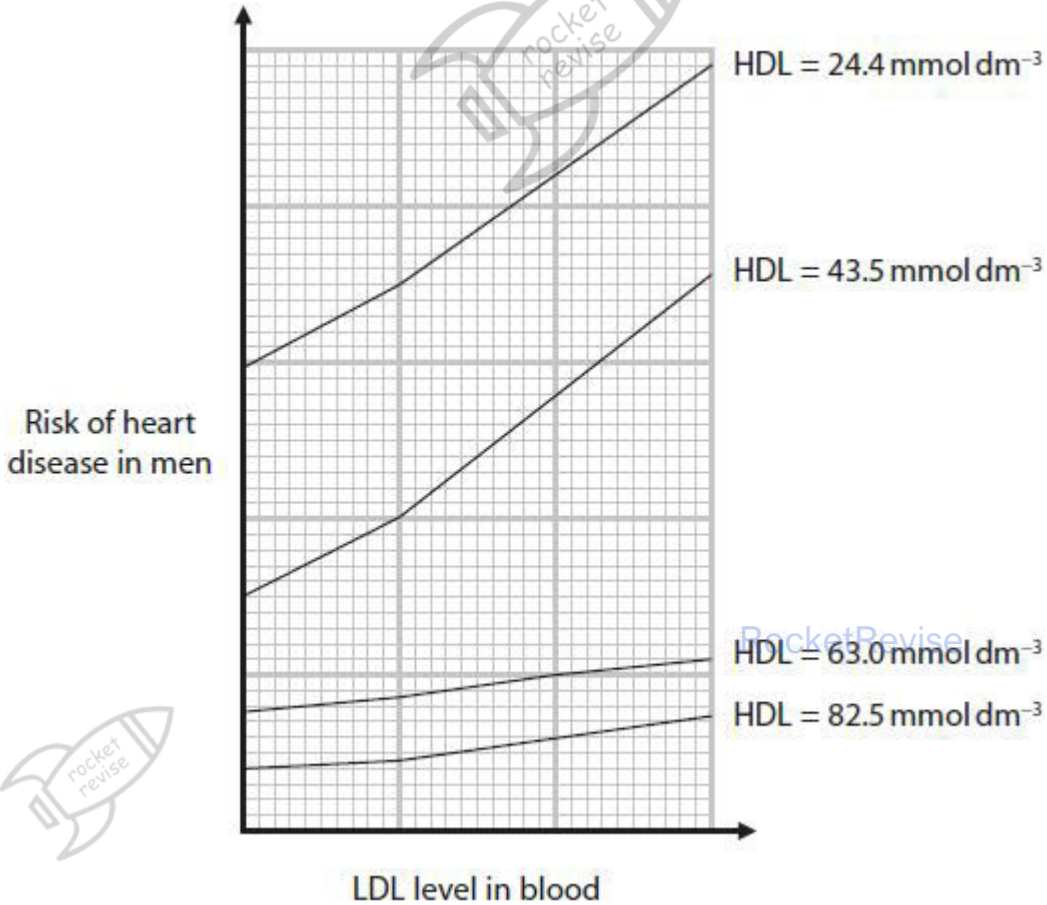
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A number of factors affect the risk of a person developing heart disease.

One factor affecting this risk is the level of high-density lipoprotein (HDL) in the blood.

* (a) The graph shows the effects of HDL levels and low-density lipoprotein (LDL) levels in the blood on the risk of men developing heart disease.



For men, a blood HDL level greater than 40 mg dm^{-3} is thought to be optimal.
Explain why a man with a blood HDL level greater than 40 mg dm^{-3} may still have a high risk of developing heart disease.
Use the information in the graph and your own knowledge to support your answer.

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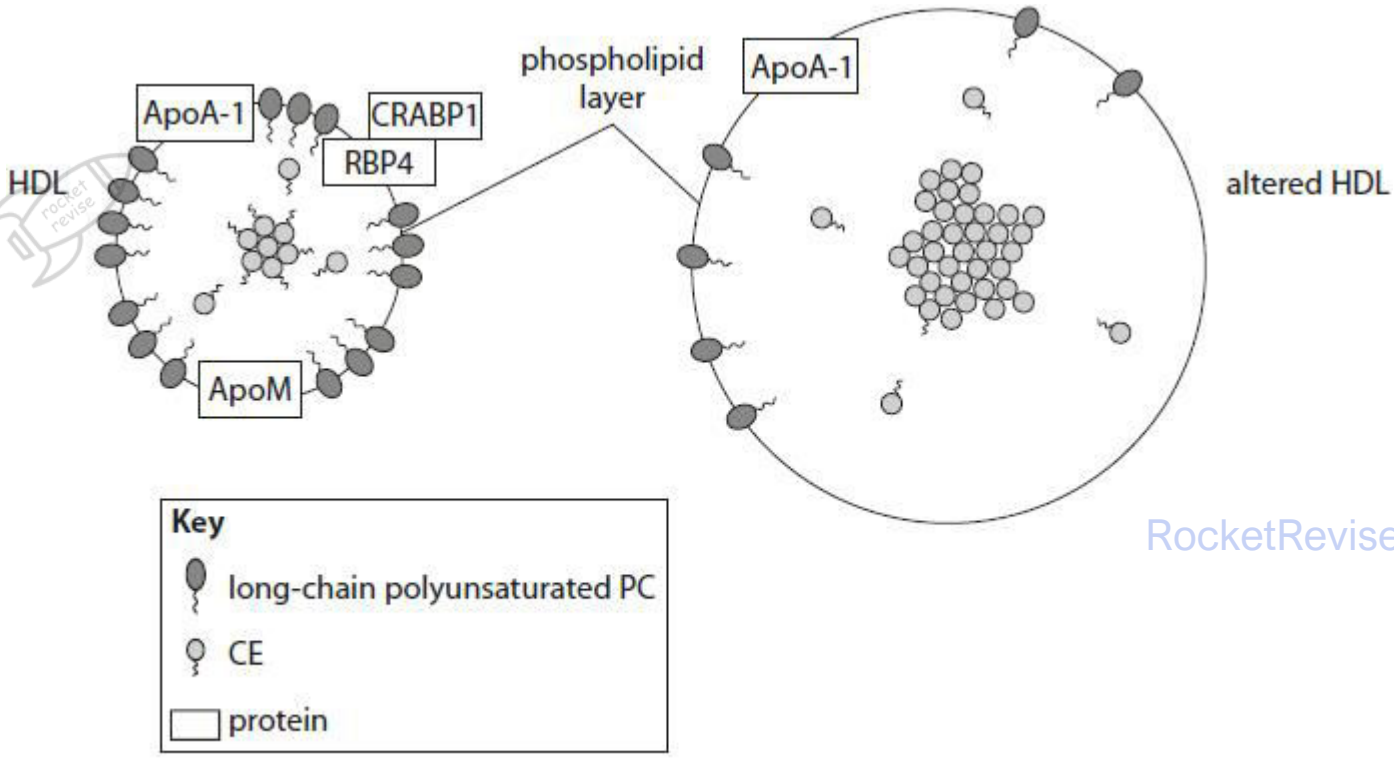
(b) Very high levels of cholesterol in the blood can alter the structure of HDL.
This altered HDL is less effective in reducing the risk of heart disease.

The diagram shows the structure of HDL in blood with a low level of cholesterol and altered HDL in blood with a high level of cholesterol.

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(Source: https://www.sciencedirect.com/science/article/pii/S0735109717373448)

(i) Compare and contrast the structure of HDL with altered HDL.

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(ii) The antioxidant properties of altered HDL are reduced.
Explain the effect that this has on reducing the risk of heart disease.

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Q10.

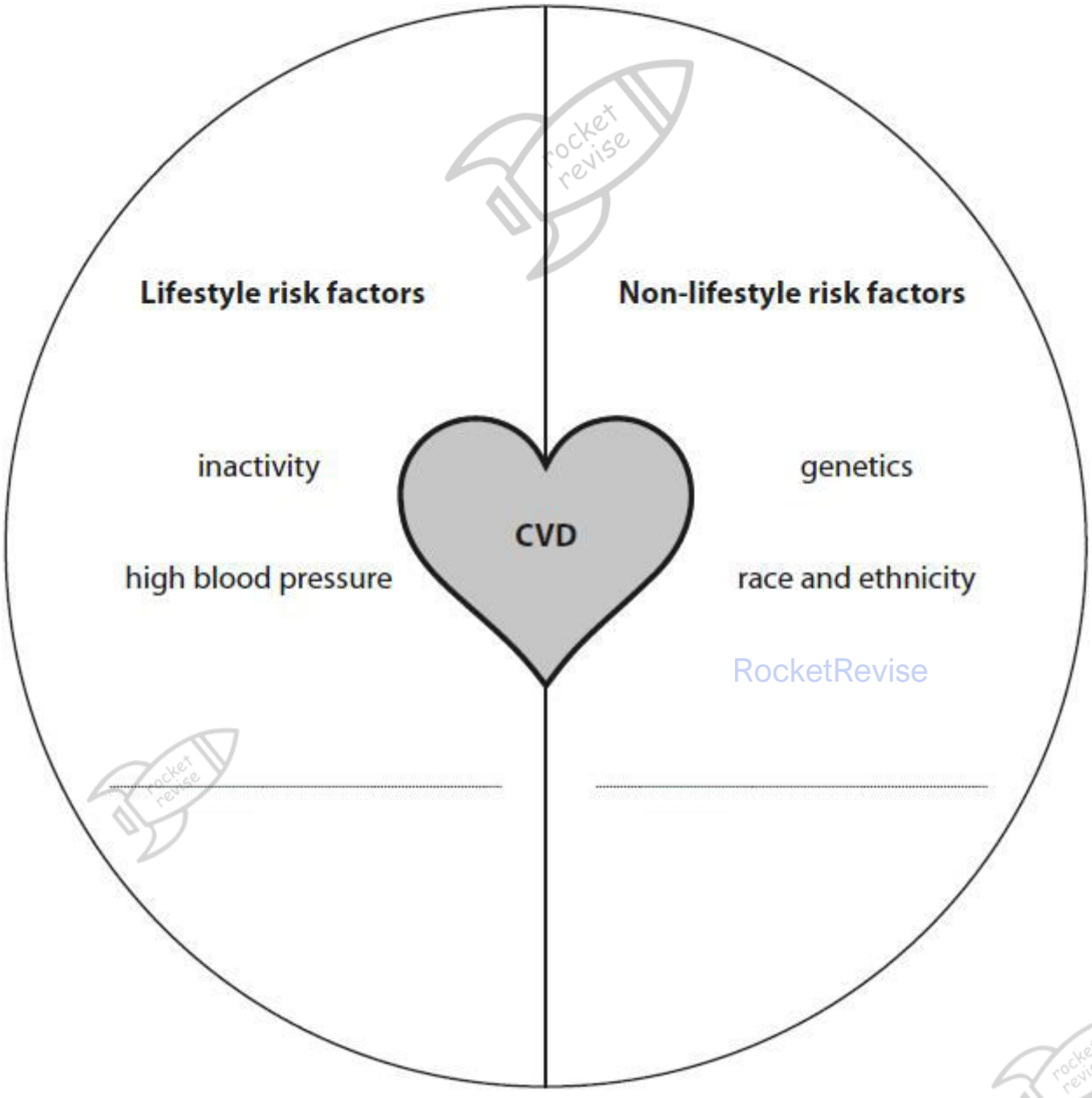
Several factors increase the risk of cardiovascular disease (CVD).

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(a) The diagram shows some factors that increase the risk of CVD.

Complete the diagram with **one** lifestyle risk factor and **one** non-lifestyle risk factor.

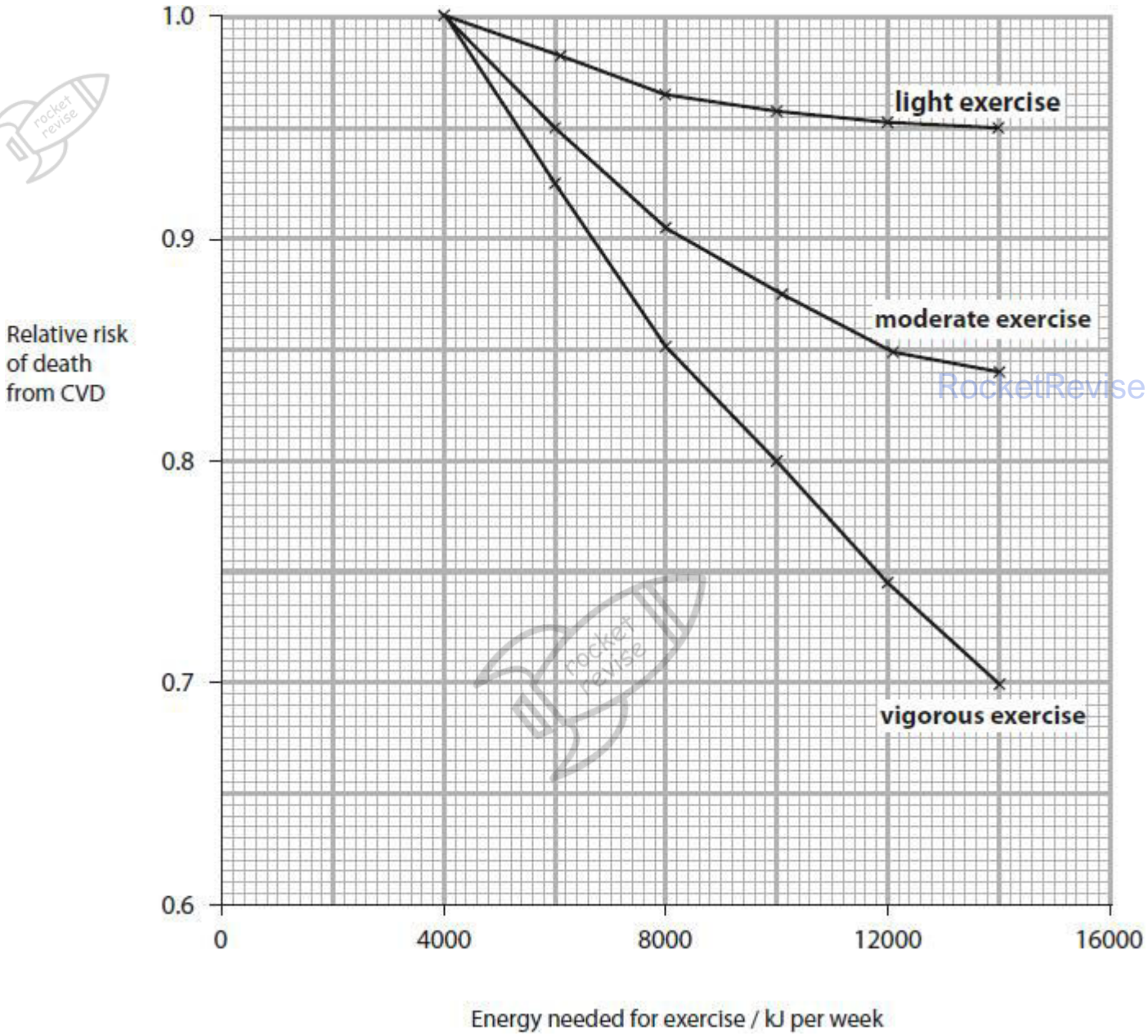
(1)



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(b) The graph shows the effect of exercise on the relative risk of death from CVD.



Describe the conclusions that can be made from the information shown in this graph.

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(c) Dietary antioxidants may reduce the risk of CVD.

(i) Explain how dietary antioxidants reduce the risk of CVD.

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(ii) Devise a study to confirm that antioxidants reduce the risk of CVD.

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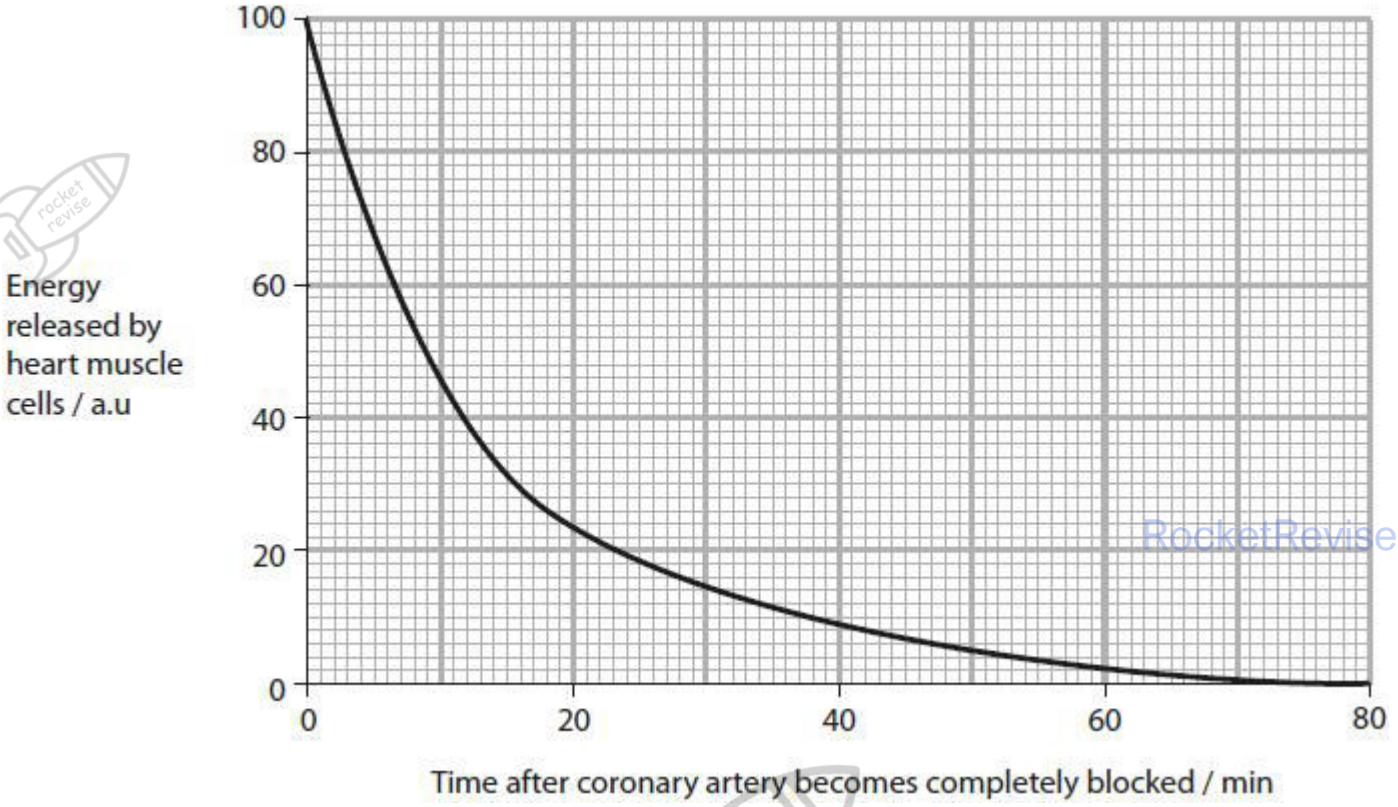
*(d) The coronary artery may become completely blocked in people with CVD.

The graph shows how the energy released by heart muscle cells changes after the coronary artery becomes completely blocked.

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The heart muscle cells no longer contract 8 minutes after the coronary artery becomes completely blocked.
The heart muscle cells begin to die 20 minutes after the coronary artery becomes completely blocked.
Explain the effects on the heart function after the coronary artery becomes completely blocked.
Use the information shown in the graph and your own knowledge to support your answer.

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(Total for question = 15 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Warfarin is an anticoagulant.

The dose of warfarin that is given to a patient has to be monitored regularly. This is done by adding thromboplastin to a blood sample and measuring the time it takes for the blood to clot.

The result is compared with a control sample and is reported as an International Normalised Ratio (INR) level.

(a) How many of the following statements are correct for thromboplastin?

- it is stored in platelets
- it is an active enzyme
- it is converted into thrombin

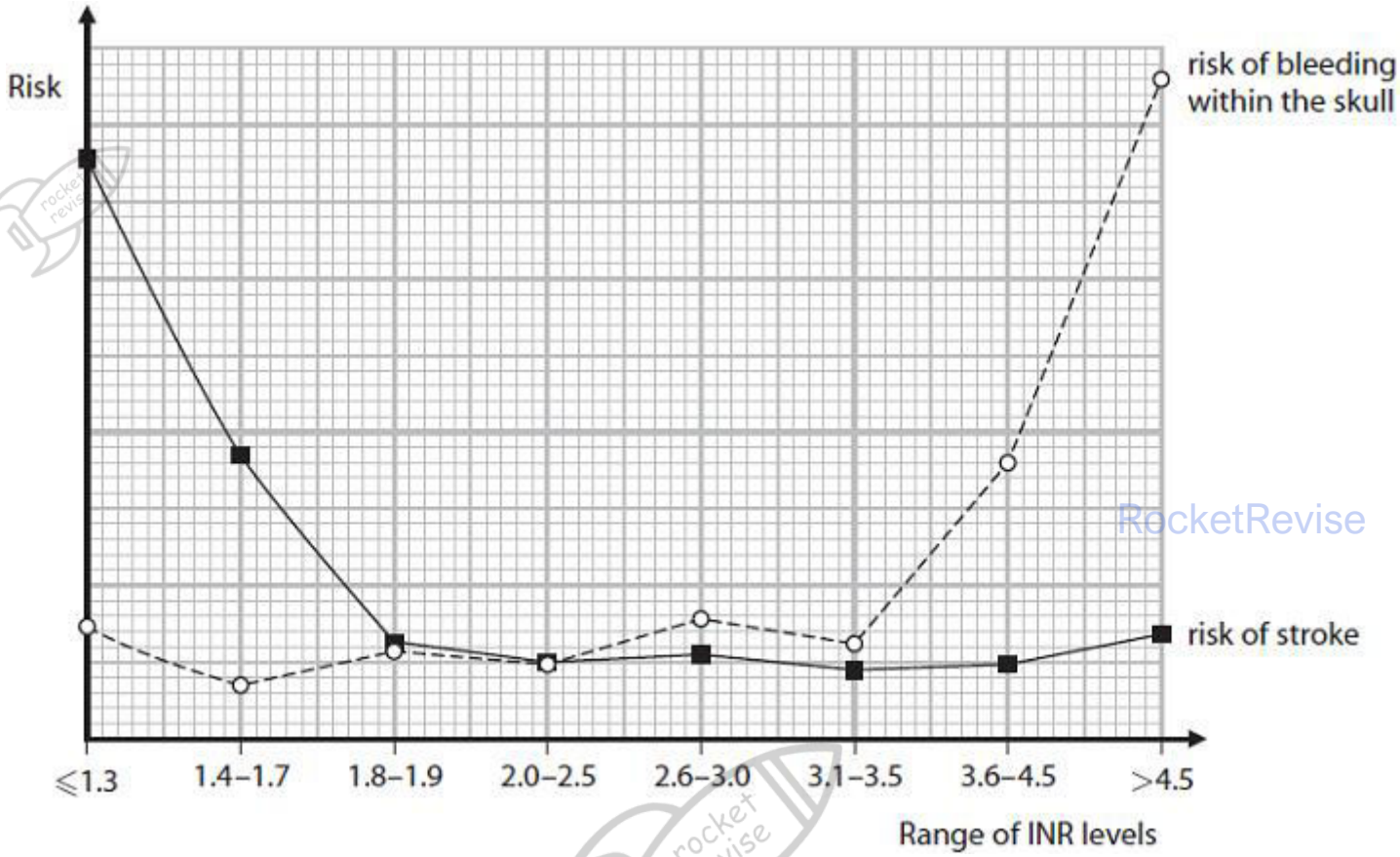
(1)

☐ **A** none
☐ **B** one
☐ **C** two
☐ **D** three

(b) The graph shows the relationship between the INR level and the risk of stroke and the risk of bleeding within the skull.

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Deduce a safe range for the INR level.
Use the information in the graph to support your answer.

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(c) Explain how taking this anticoagulant can reduce the risk of cardiovascular disease (CVD).

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(d) Warfarin is given to patients who have had a heart attack, to prevent further heart attacks and other related health issues.

A study compared warfarin with the use of aspirin, over a four-year period following the first heart attack.

(i) A group of 1216 were given warfarin; 83.3% of these people had no further health issues.
Calculate the number of people given warfarin who had no further health issues following their first heart attack.

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(1)

Answer .....

(ii) A second study compared this result with people given a combination of warfarin and aspirin.
Describe how this study should be carried out to produce a valid comparison.

(4)

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(Total for question = 12 marks)

Q12.

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Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The risk of developing cardiovascular disease (CVD) is affected by two groups of factors:

- lifestyle factors that can be changed
- non-lifestyle factors that cannot be changed.

Methods to reduce the risk of developing CVD include drug treatments and lifestyle changes.

(a) (i) Which row of the table identifies one lifestyle factor and one non-lifestyle factor?

(1)

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		Lifestyle factor	Non-lifestyle factor
☐	**A**	body mass index (BMI)	age
☐	**B**	gender	high alcohol intake
☐	**C**	genetics	high blood pressure
☐	**D**	high blood cholesterol	inactivity

(ii) Explain why a person might have to take several types of drugs to reduce the risk of CVD.

(2)

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(b) Nutritional studies have shown that dietary antioxidants can reduce the risk of CVD.

(i) Explain why antioxidants in the diet reduce the risk of CVD.

(3)

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(ii) Some studies do not assess the nutritional quality of the diet of the participants.

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Explain why the results of these studies have to be treated with caution.

(3)

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(iii) Explain why changes in diet, other than antioxidants, can reduce the risk of CVD.

(2)

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(Total for question = 11 marks)

Q13.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

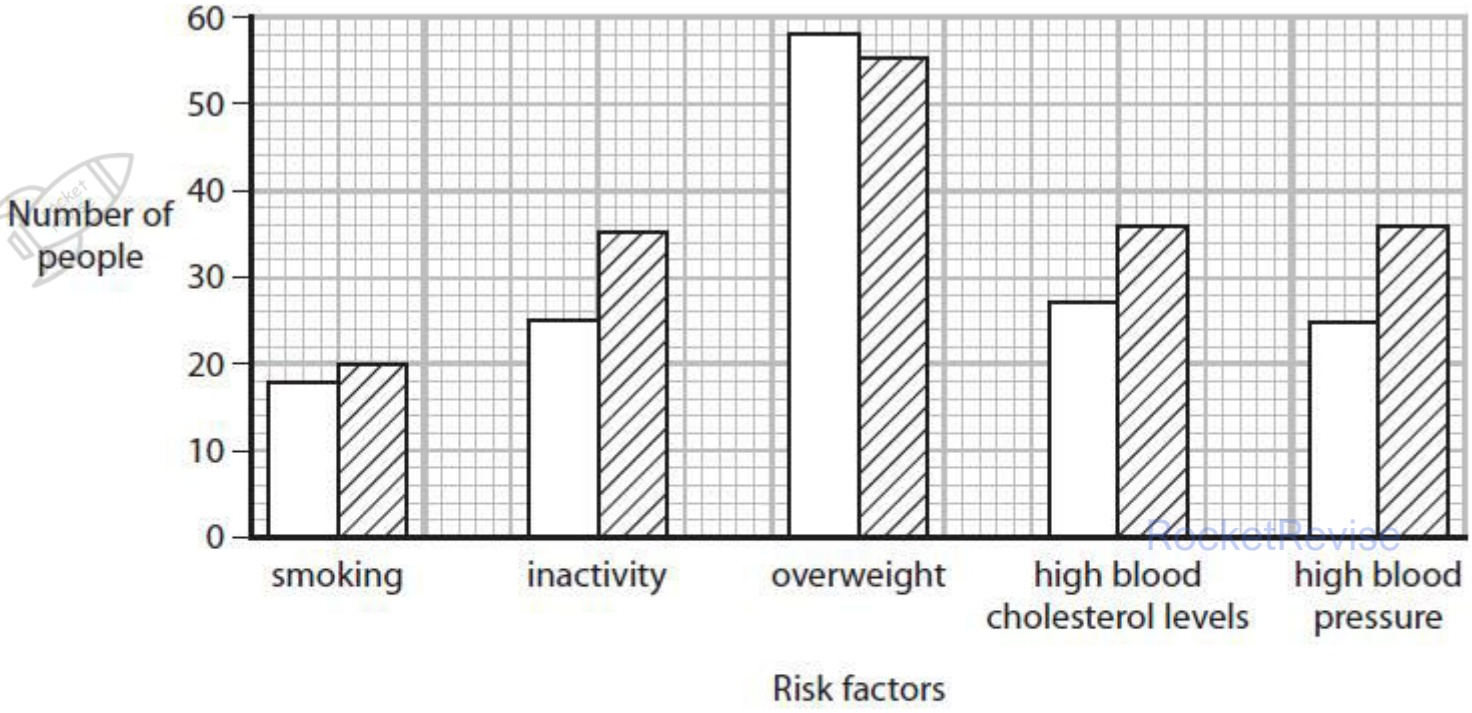
The risk of developing cardiovascular disease (CVD) is dependent on a number of factors.

(a) People's perception of these risks are often very different from the actual risks.

A study was designed to compare perceived risks with actual risks.
A group of people were asked if they thought they were at risk (perceived risk) of CVD. The same group of people were then examined to determine their actual risks of CVD.
The graph shows the results of this study.

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(i) Describe **two** conclusions that can be drawn from this study.

(2)

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(ii) Explain how this study should have been designed to allow valid conclusions to be made.

(3)

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(iii) Explain why it is important that a person's perception of a risk is close to the actual risk.

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(b) The 10-year risk of a person developing CVD can be estimated in a number of ways.

The Framingham Risk Score uses a point scoring system to estimate the 10-year risk of developing CVD.

Table 1 shows the point scores for different risk factors in women.

Table 1

Risk factor	Points per age group			
	35 to 39	40 to 44	45 to 49	50 to 54
age	−3	0	+3	+6
total cholesterol level / mg dm^{-3}				
<160	0	0	0	0
160 to 199	+4	+3	+3	+2
200 to 239	+8	+6	+6	+4
240 to 279	+11	+8	+8	+5
smoker	+9	+7	+7	+4
systolic blood pressure / mm Hg				
<120	0	0	0	0
120 to 129	+1	+1	+1	+1
130 to 139	+2	+2	+2	+2

Table 2 shows the 10-year risk for the total points scored, calculated using the information in Table 1.

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Table 2

Total points scored	10-year risk of developing CVD (%)
16	4
17	5
18	6
19	8
20	11
21	14
22	17

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(i) Calculate the 10-year risk of CVD for a 35-year-old woman who:

- has a total cholesterol level of 242 mg dm^{-3}
- is a smoker
- has a systolic blood pressure of 132 mm Hg. **(2)**

Answer

..... %

(ii) Explain the lifestyle changes this woman could make to reduce her 10-year risk of CVD. Use the information in the table to support your answer.

(2)

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(iii) Which pair of drugs may reduce the 10-year risk of CVD for this woman?

(1)

- ☒ **A** anticoagulants and statins
- ☒ **B** anticoagulants and antihypertensives
- ☒ **C** antihypertensives and platelet inhibitors
- ☒ **D** antihypertensives and statins

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(iv) Suggest why this method of determining the 10-year risk of CVD may not be accurate.

(2)

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(Total for question = 15 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The Portfolio diet is a diet that has been shown to lower low-density lipoproteins (LDLs) and other risk factors of cardiovascular disease (CVD).

(a) (i) Which row in the table describes the cholesterol content and antioxidant content of this diet?

(1)

		Cholesterol content	Antioxidant content
☒	**A**	high	high
☒	**B**	high	low
☒	**C**	low	high
☒	**D**	low	low

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(ii) Name **two** other dietary risk factors that could be changed in this diet.

(1)

1 .....

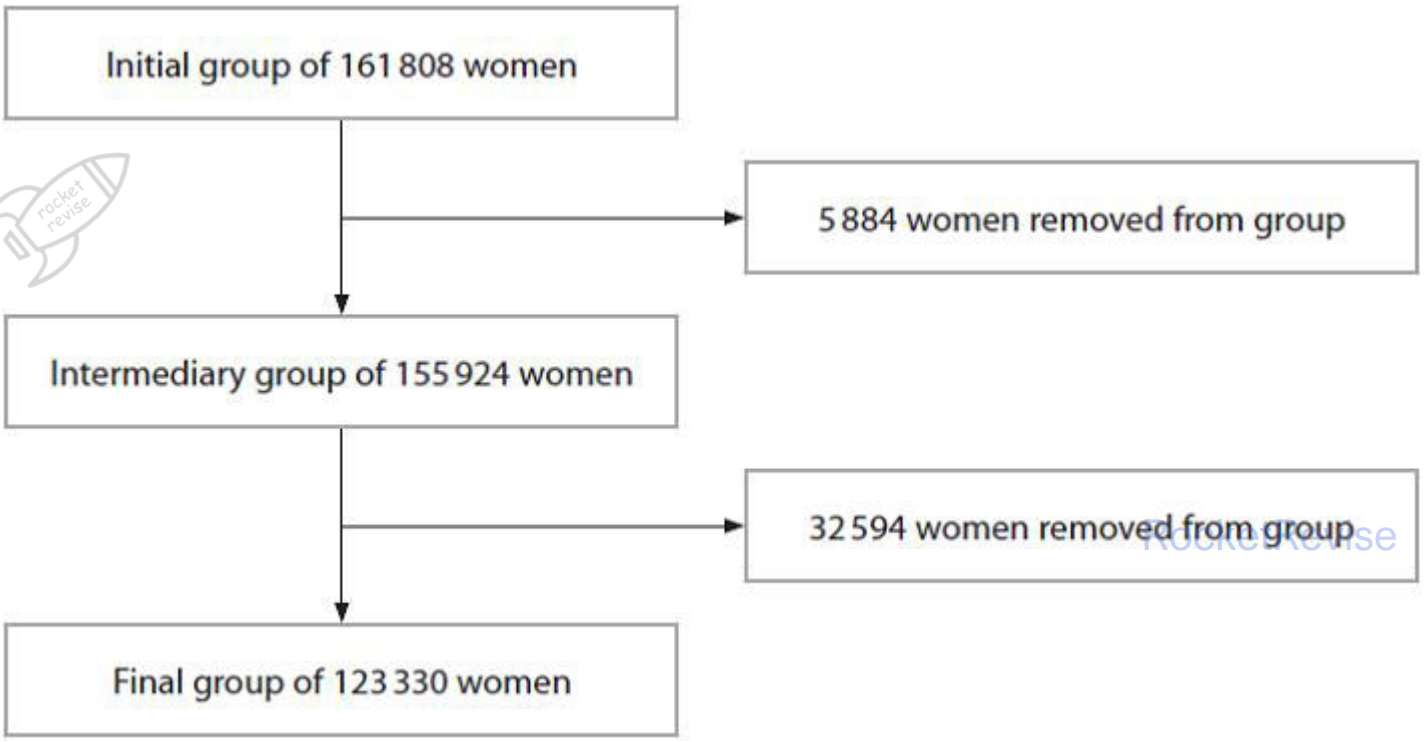
2 .....

(b) The effect of this diet on the development of CVD in women was investigated.

The diagram shows how 123 330 women were selected from an initial group of 161 808 women.

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(i) Some of the women were removed from the initial group because there was information missing about their lifestyles.
Suggest why women with missing lifestyle information were removed from the initial group.

(1)

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(ii) Some of the women were removed from the initial group because they gave unrealistic estimates of their daily energy intake.
Give **two** reasons why some women may have given unrealistic estimates of their daily energy intake.

(2)

1 .....

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2 .....

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(iii) The women removed from the intermediary group had signs of CVD.
Explain why these women were removed from this investigation.

(2)

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(iv) Calculate the percentage of all the women removed from this investigation.

(1)

Answer ..... %

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(c) Explain why a diet that is low in saturated fat is likely to reduce the risk of CVD.

(3)

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(Total for question = 11 marks)

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Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The risk of developing cardiovascular disease (CVD) is influenced by a number of factors.

Risk assessment calculators allow a person to calculate their own risk of developing CVD.

The person completes a questionnaire about their health and lifestyle. This information is then used to calculate the risk of the person developing CVD within the next ten years.

The table shows the information required by two different risk assessment calculators, RAC-1 and RAC-2.

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Information	Information required by the calculators	
	RAC-1	RAC-2
Age	✓	✓
Blood pressure	✓	✓
Smoking	✓	x
Total cholesterol	✓	✓
HDL cholesterol	x	✓

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(a) How many of the risk factors listed in the table are **not** influenced by lifestyle?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

*(b) Explain the effectiveness of these two calculators in the assessment of risk and whether other factors should be considered.

(6)

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(c) The calculators work out the risk of a person developing CVD in the next 10 years.

(i) Explain why the information entered by a person may lead to an underestimate of their 10-year risk of developing CVD.

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(2)

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(ii) Suggest why these two risk calculators may not be suitable for everybody to use.

(2)

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(iii) Suggest why these risk calculators estimate the **10-year** risk of developing CVD.

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(1)

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(Total for question = 12 marks)

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Q16.

In 2021, it was reported that cardiovascular diseases (CVDs) kill 17.9 million people per year. This is 31% of all global deaths.

(a) (i) Calculate how many global deaths there are in one year.

Express your answer in standard form.

(2)

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Answer .....

(ii) It is predicted that by 2030 more than 22.2 million people will die each year from CVDs.
Suggest how this prediction was made.

(2)

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(b) The rhizomes of a ginger plant are used in cooking.

The photograph shows the rhizomes of a ginger plant.



(Source: © Helen Sessions / Alamy Stock Photo)

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These rhizomes have high levels of antioxidants.
Some people believe that eating these rhizomes reduces the risk of CVD.
(i) Explain why eating these rhizomes may help to prevent CVD.

(3)

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(ii) There is very little data available to show that these rhizomes reduce the risk of CVD. Suggest why these rhizomes can be eaten to prevent CVD even though their effectiveness has not been thoroughly tested by scientists.

(1)

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*(iii) Explain how a study should be designed to collect valid and repeatable data to show the effectiveness of using ginger in preventing CVD.

(6)

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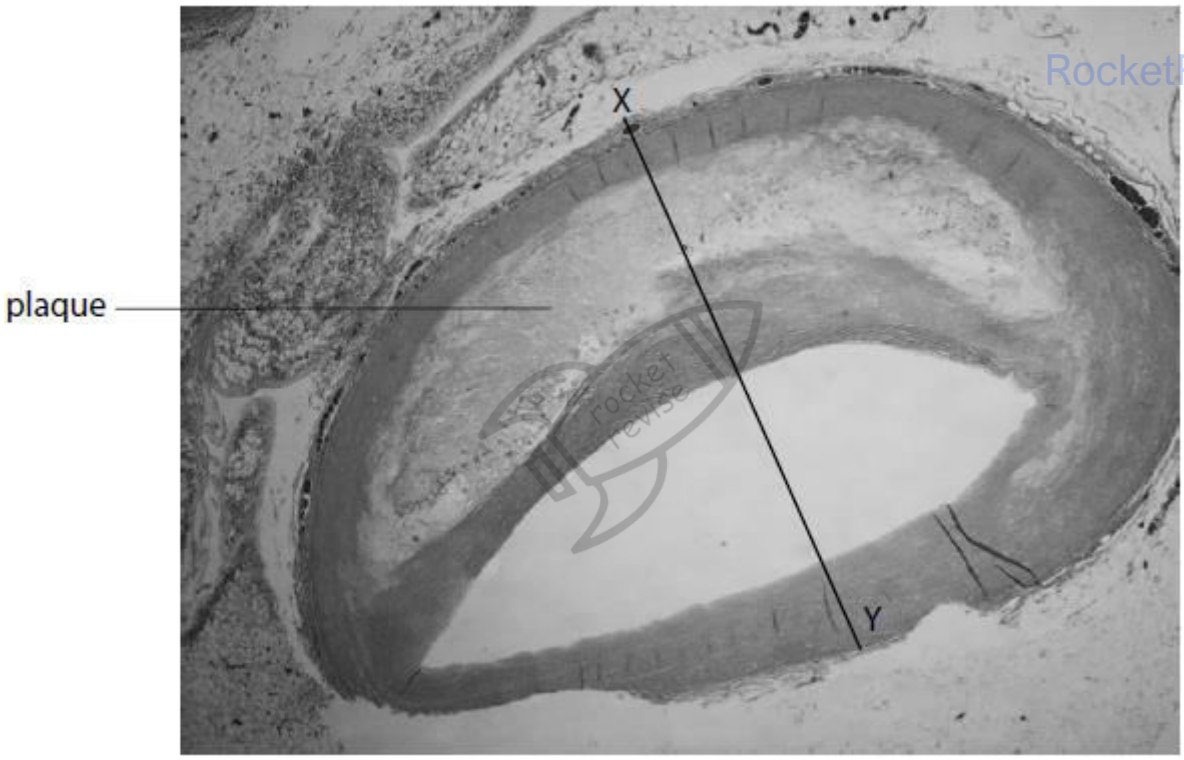
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(Total for question = 14 marks)

Q17.

Many studies have linked the development of atherosclerosis with cardiovascular disease (CVD).

(a) The photograph shows a section through an artery with a plaque (atheroma) from a patient with atherosclerosis.



© Ogphoto/iStock

Calculate the percentage increase in the thickness of the artery wall where the plaque is located. Take your measurements along the line labelled X–Y.

(3)

Answer ..... %

(b) Explain how atherosclerosis can result in damage to the heart muscle.

(4)

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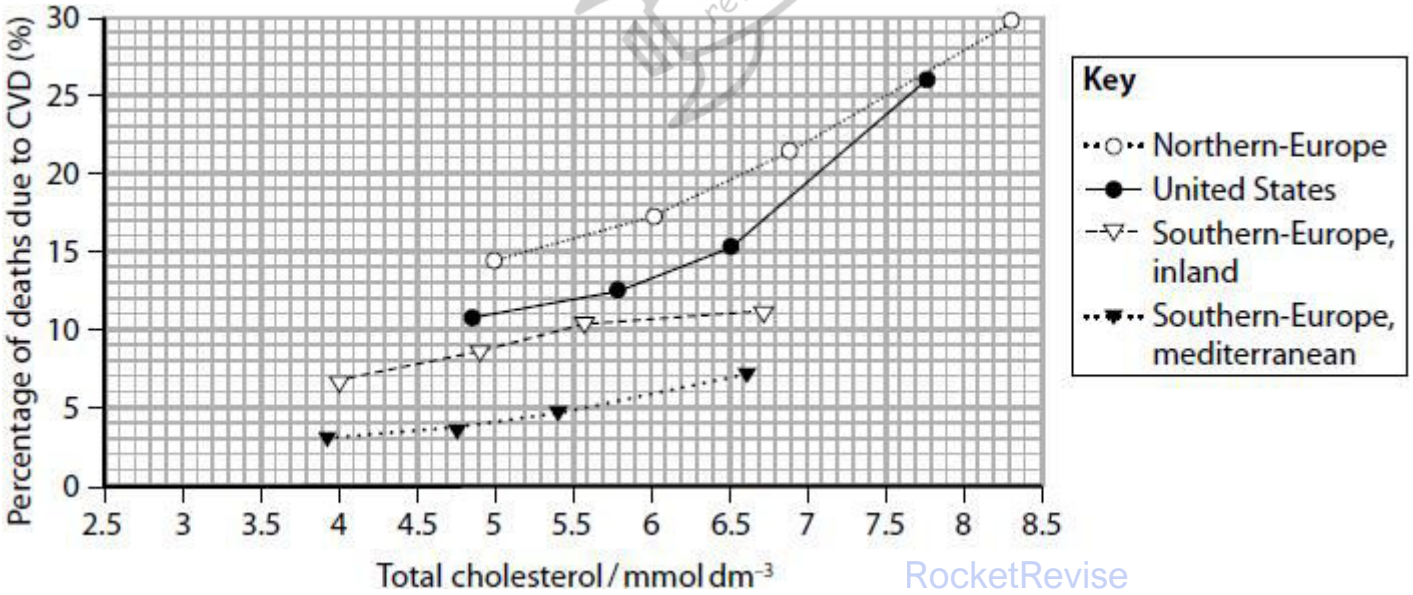
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*(c) Cholesterol is transported in the blood as lipoproteins LDL and HDL.

In one study, the relationship between total blood cholesterol and the risk of death from CVD was investigated.
The results are shown in the graph.



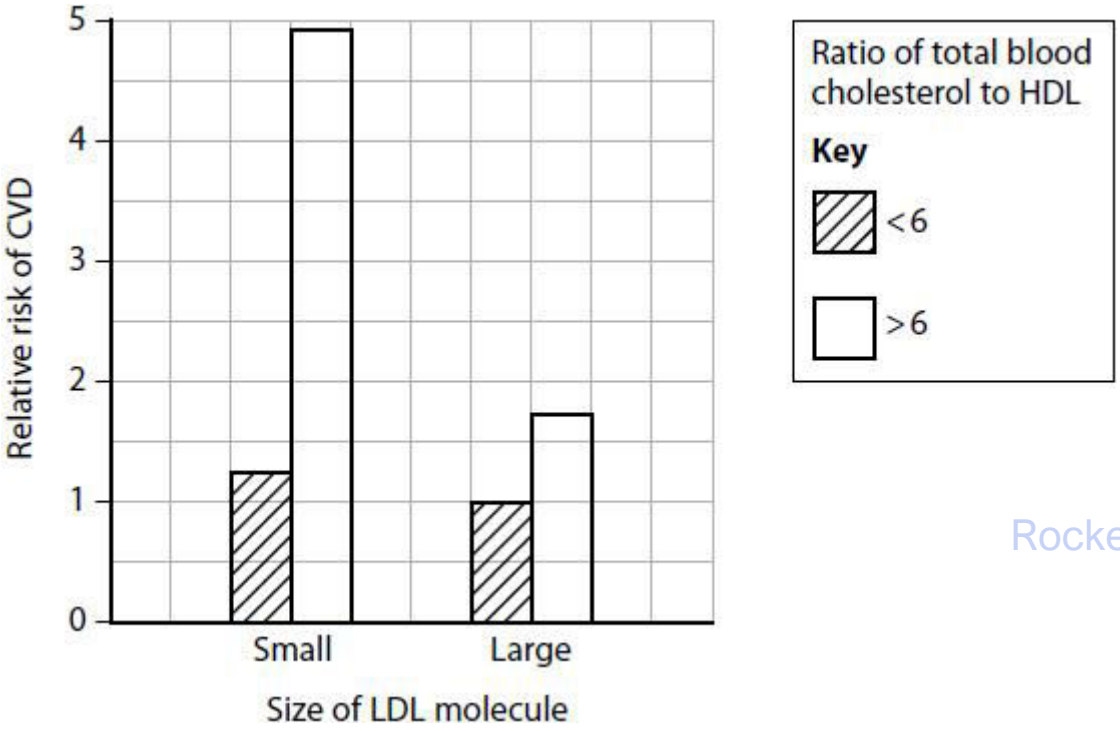
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Graph 1

In another study, the effect of the size of LDL and the ratio of total blood cholesterol to HDL on the relative risk of CVD was investigated.
The results are shown in the graph.

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Graph 2

Assess the contribution of lipoproteins to the risk of developing CVD. Use the information in the graph to support your answer.

(6)

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(Total for question = 13 marks)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows two avocados.

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© Andrii Gorulko / Alamy Stock Photo

(a) Avocados are high in unsaturated fatty acids and low in saturated fatty acids.

Compare and contrast the structure of an unsaturated fatty acid and the structure of a saturated fatty acid.

(3)

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(b) The effect of eating avocados each week on the risk of coronary heart disease (CHD) was investigated.

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A group of men and women who had no signs of cardiovascular disease (CVD) or stroke were selected. They were all between the ages of 30 and 75.
The group ate avocados each week to replace animal products in their diet whenever possible.
The group all completed a questionnaire about the food they ate at the start of the investigation and every four years after that for over 30 years.

(i) There were 41 701 men and 68 786 women in the group.
Which is the ratio of men to women?

(1)

- ☒ **A** 0.6:1
- ☒ **B** 1:0.6
- ☒ **C** 1:2
- ☒ **D** 2:1

*(ii) Describe the questions that need to be in the questionnaire so that appropriate information can be gathered for this investigation.
Give reasons for choosing these questions.

(6)

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(iii) Comment on the validity of the methods used in this investigation.

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(Total for question = 14 marks)

Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Several factors are associated with the risk of developing cardiovascular disease (CVD).

Obesity is one risk factor associated with CVD.

The ability of a group of people to lose weight was investigated.

This investigation considered:

- the body mass index (BMI) of the person
- the levels of amylase in their saliva
- the bacteria present in their digestive system.

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(a) The BMI of a person can be calculated using the formula:

BMI = mass in kg ÷ (height in m)^2

(i) Name **one** other obesity indicator.

(1)

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(ii) Which of the following is the BMI of a person who has a mass of 65 kg and a height of 165 cm?

(1)

☒ **A** 23
☒ **B** 24
☒ **C** 39
☒ **D** 40

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(iii) A person's BMI is used to put them into a weight category
A person with a BMI of 25 and above, but below 30, is considered to be overweight but not obese.
Which describes the body mass of a person who is obese?

(1)

- ☒ **A** BMI > 30
- ☒ **B** BMI ≥ 30
- ☒ **C** BMI < 30
- ☒ **D** BMI ≤ 30

(b) People who are overweight or obese are encouraged to lose weight.

State **two** ways in which a person could lose weight.

(1)

1 .....

2 .....

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(c) Amylase in saliva breaks down amylose into maltose.

(i) Which row of the table describes the action of amylase?

(1)

		Type of reaction	Glycosidic bond broken
☒	**A**	condensation	1 – 4
☒	**B**	condensation	1 – 6
☒	**C**	hydrolysis	1 – 4
☒	**D**	hydrolysis	1 – 6

(ii) People with low levels of amylase in their saliva were found to lose weight more easily.
Suggest why these people lost weight more easily.

(2)

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(d) The bacteria in people's digestive systems replicate.

The investigation found that people with greater weight loss had bacteria with increased rates of replication.

(i) Explain why this finding can be described as a positive correlation.

(2)

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(ii) Suggest why the rate of replication of the bacteria could affect weight loss.

(2)

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(Total for question = 11 marks)

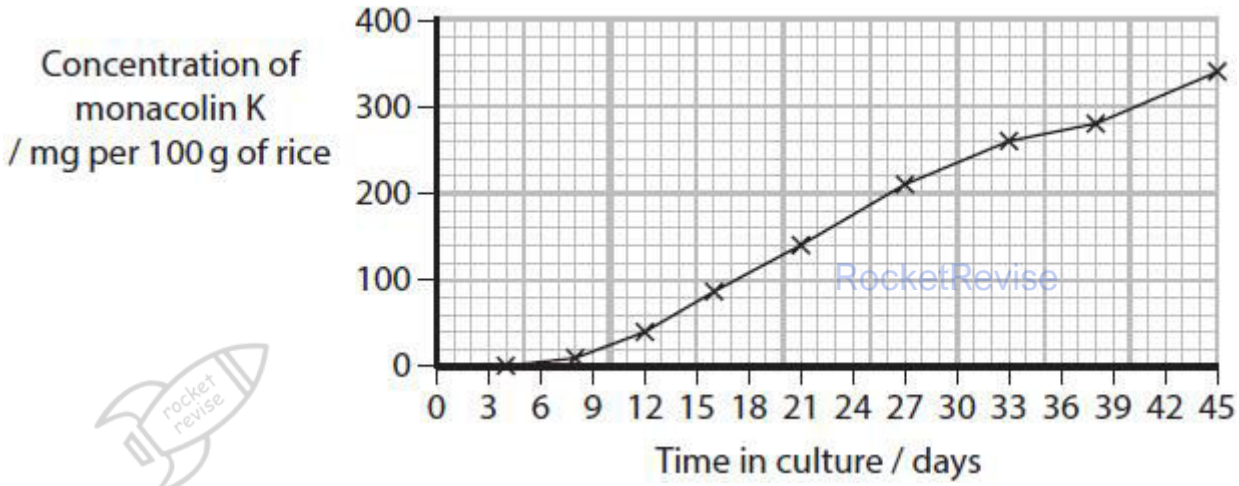
Q20.

Statins are used in the prevention and treatment of cardiovascular disease (CVD).

Red yeast rice (RYR) is a source of food and medicine in Asia. It is produced by the fermentation of rice by yeast.

Red yeast rice contains monacolin K, which is similar to a statin.

(a) The graph shows the production of monacolin K during the fermentation of rice by yeast.



Calculate the mean rate of production of monacolin K in this culture, in the 45-day time period.
Give your answer to **two** decimal places.

(2)

Answer ..... mg per 100 g of rice per day

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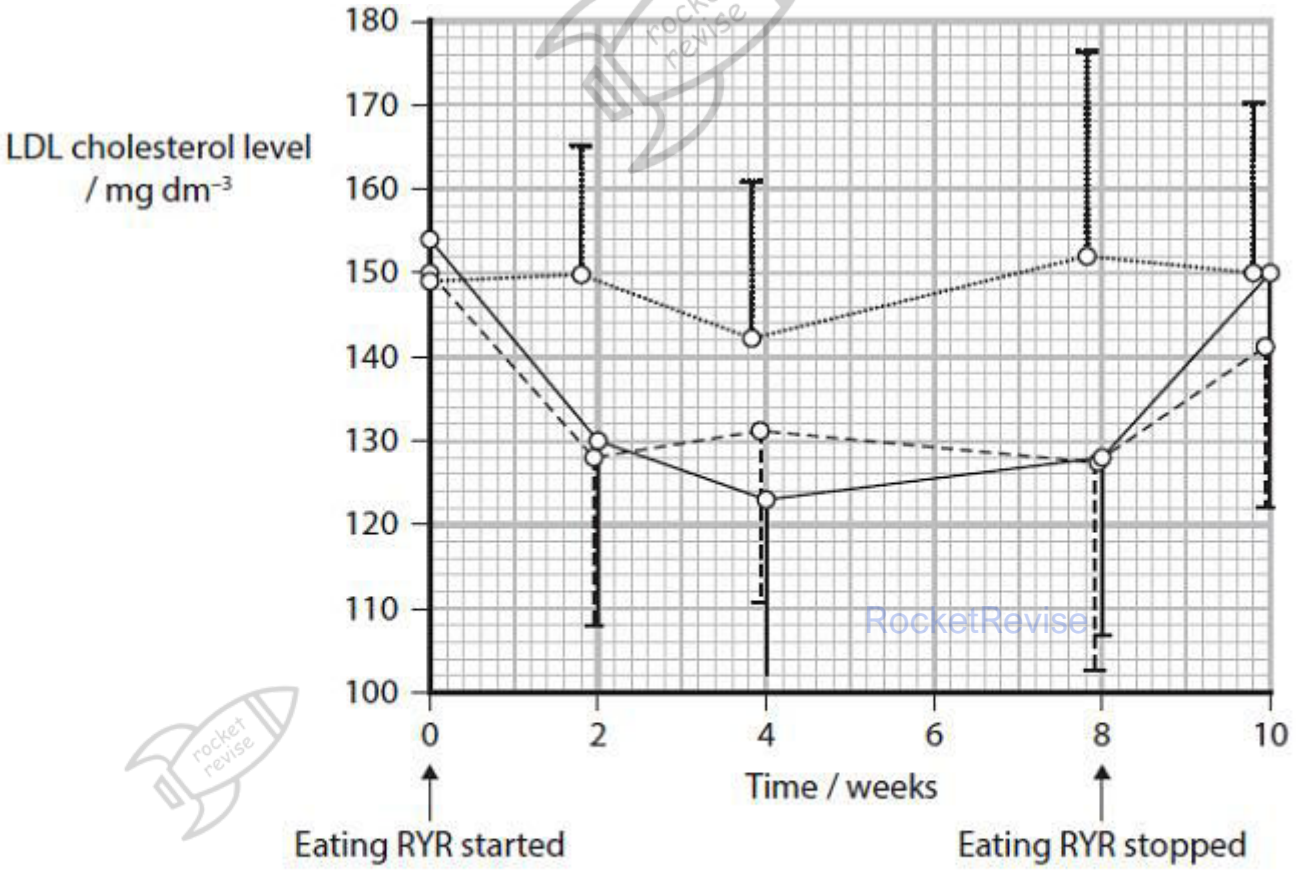
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*(b) Scientists have studied the effects of RYR on lipid levels.

The table shows the results of four studies on the effects of eating RYR on lipid levels.

Study	Mass of RYR / mg per day	Percentage change in lipid levels (%)		
		Decrease in total cholesterol	Decrease in LDL cholesterol	Change in HDL cholesterol
1	340	16 to 17	21 to 24	Between a decrease of 3 and an increase of 3
2	1200	20	34	Increase of 18
3	2400	13	23	No data
4	1200	13	23	No data
	2400	24	32	No data

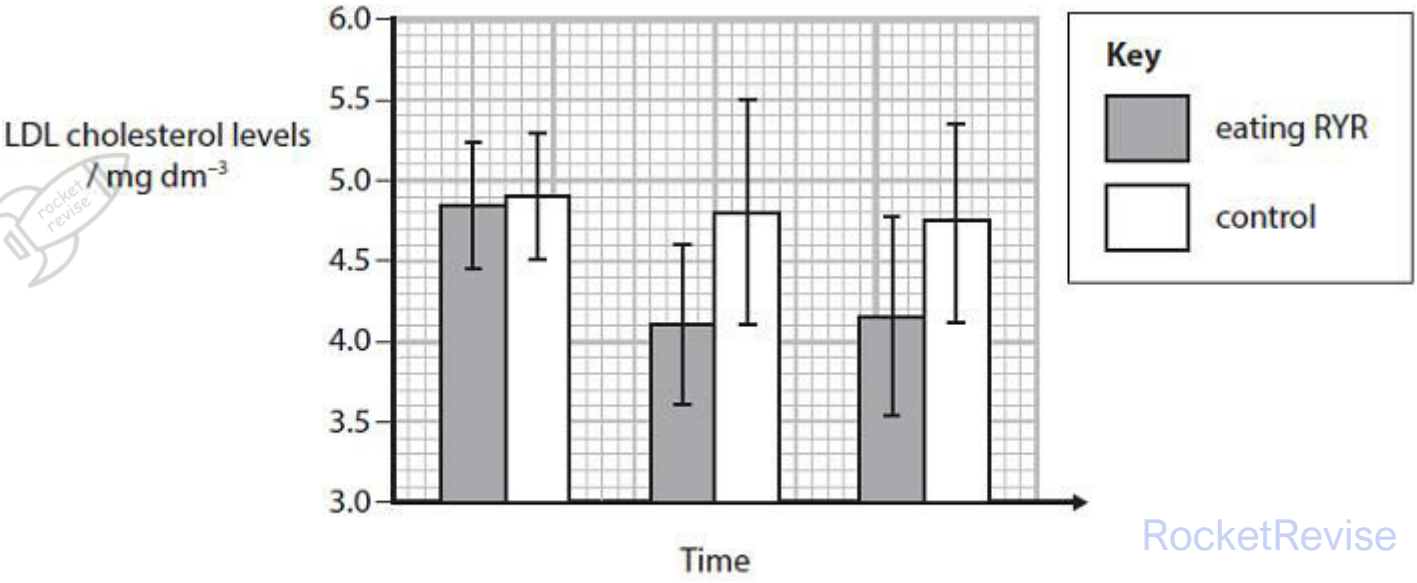
Graph 1 shows the effects of eating RYR daily on LDL cholesterol levels.
It also shows the changes in levels of LDL cholesterol when eating RYR was stopped.
The investigation included a control group that did not eat RYR.



Graph 2 shows the results of another study where the effect of eating RYR on LDL cholesterol levels was compared with a control group.

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Graph 2

Discuss the limitations of using the results of these studies to draw conclusions about the effects of RYR on lipid levels.

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(c) Treatment with statins can have side effects.

(i) Name **two** possible side effects of being treated with statins.

(1)

1 .....

2 .....

(ii) Describe how an investigation could be designed to identify any side effects of treatment with monacolin K.

(4)

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(iii) Suggest **two** problems when assessing possible side effects resulting from treatment with monacolin K in an investigation.

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(Total for question = 15 marks)

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Q21.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Bears hibernate during the winter months, so are inactive for long periods of time.

The photograph shows two bears hibernating.



(Source: © Minden Pictures / Alamy Stock Photo)

Hibernating bears do not develop deep vein thrombosis (DVT).

Deep vein thrombosis is a blood clot that forms within a deep vein, often in the leg.

In humans, short-term periods of inactivity, such as sitting in an aeroplane, increase the risk of DVT.

This is due to reduced blood flow in the veins, causing damage to the walls of the veins.

(a) (i) Which molecule, released from platelets, results in blood clotting?

(1)

- ☐ **A** fibrinogen
- ☐ **B** prothrombin
- ☐ **C** thrombin
- ☐ **D** thromboplastin

(ii) Which pair of molecules are enzymes in their active form?

(1)

- ☐ **A** fibrinogen and prothrombin
- ☐ **B** prothrombin and thrombin
- ☐ **C** thrombin and thromboplastin
- ☐ **D** thromboplastin and prothrombin

(iii) Which molecule found in the blood is insoluble?

(1)

- ☐ **A** fibrin
- ☐ **B** fibrinogen
- ☐ **C** prothrombin

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☒ **D** thromboplastin

(b) Pulmonary embolism can be caused by a DVT when the blood clot travels through the circulatory system and blocks the pulmonary artery.

Symptoms of pulmonary embolism include difficulty in breathing and a rapid heart rate.
Suggest why pulmonary embolism can result in these symptoms.

(3)

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(c) A study of bears in Sweden discovered that these bears produced less HSP47 protein when hibernating in the winter.

This protein is found on platelets and is involved in the binding of platelets to damaged blood vessels.
Discuss whether this discovery could be useful to humans.

(3)

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(Total for question = 9 marks)

Q22.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

One risk factor for cardiovascular disease is a short blood clotting time.

One test for measuring blood clotting time is the prothrombin time (PT).

(a) Read through the following passage about prothrombin and the blood clotting process.

Complete the passage by writing the most appropriate word or words on the dotted lines.

(3)

Prothrombin is transported in the .....of a human.
It is converted into thrombin by an enzyme called
.....
The substrate for thrombin is .....

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(b) Which pair of drugs is used to treat patients who have short blood clotting times?

(1)

- ☒ **A** anticoagulants and antihypertensives
- ☒ **B** anticoagulants and platelet inhibitors
- ☒ **C** antihypertensives and statins
- ☒ **D** platelet inhibitors and statins

(c) The range for a healthy PT is between 10 and 13 seconds.

A person with a PT of 8.5 seconds was treated with a drug to increase the PT time to 10 seconds.
Calculate the percentage increase in PT from 8.5 seconds to 10 seconds.

(1)

Answer ..... %

(Total for question = 5 marks)

Q23.

Gum disease affects up to 90% of the world's adult population.

Gum disease has been shown to be associated with the development of cardiovascular disease (CVD).

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(a) Towards the end of 2023, the world's population was 8 078 799 080.

Approximately 70% of this population were adults.
Estimate the number of adults in the world.
Give your answer in standard form.

(2)

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Answer .....

(b) The development of gum disease can be reduced by good oral hygiene. This is one example of a healthy lifestyle.

Give **two** other examples of a healthy lifestyle that can help reduce the risk of CVD.

(1)

1 .....

2 .....

(c) Gum disease is caused by inflammation.

Chemicals released into the blood as a result of the inflammation are thought to cause damage to the circulatory system.
The number of white blood cells present in the saliva increases in the early stages of gum disease.
In a study, Canadian scientists determined the correlation between the number of neutrophils in saliva and damage to the circulatory system.
Neutrophils are a type of white blood cell.
The scientists took samples of saliva from volunteers in the study, determined the blood pressure in their arteries and assessed the damage to their endothelial cells.

(i) Give the meaning of the term **correlation.**

(1)

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(ii) It was important that all the people in the study had similar blood pressure that was within a healthy range.
Explain why this was important.

(3)

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(iii) Calculate the blood pressure of a person who has a systolic blood pressure (SBP) of 11.0 kPa and a diastolic blood pressure (DBP) of 6.7 kPa, using the formula:

$$\text{blood pressure} = \frac{\text{SBP} + (2 \times \text{DBP})}{3}$$

(1)

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Answer .....kPa

(iv) The scientists assessed the damage to endothelial cells.
Explain why damage to endothelial cells might be an indication that a person is at risk of CVD.

(3)

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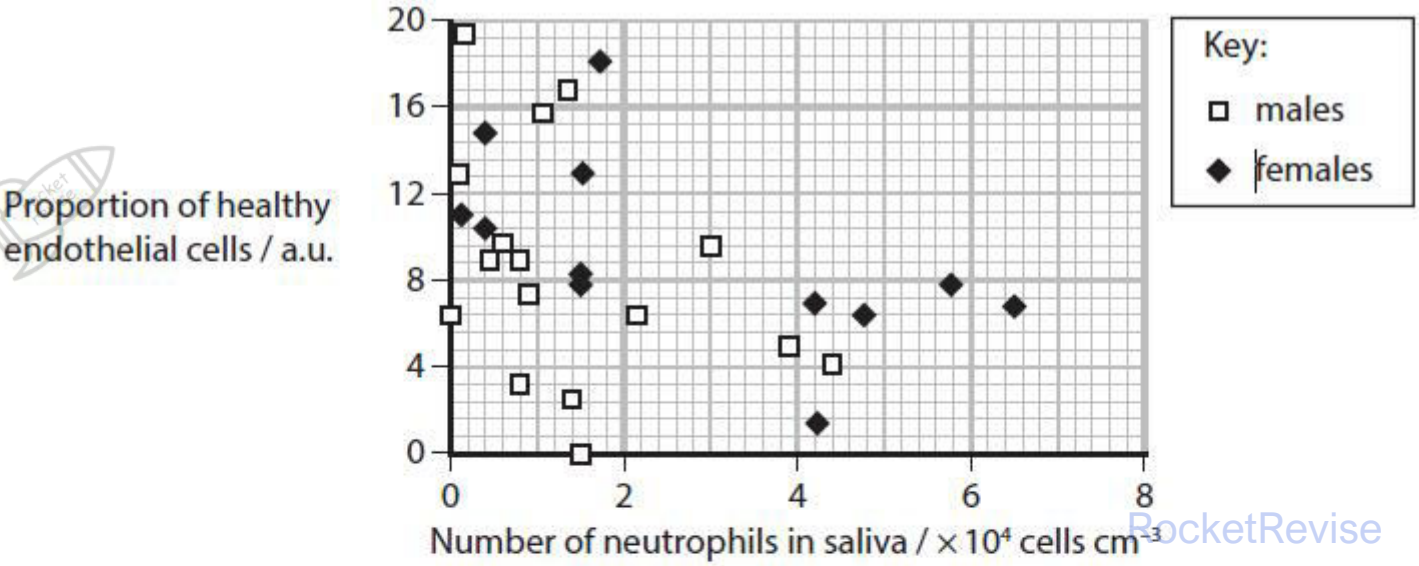
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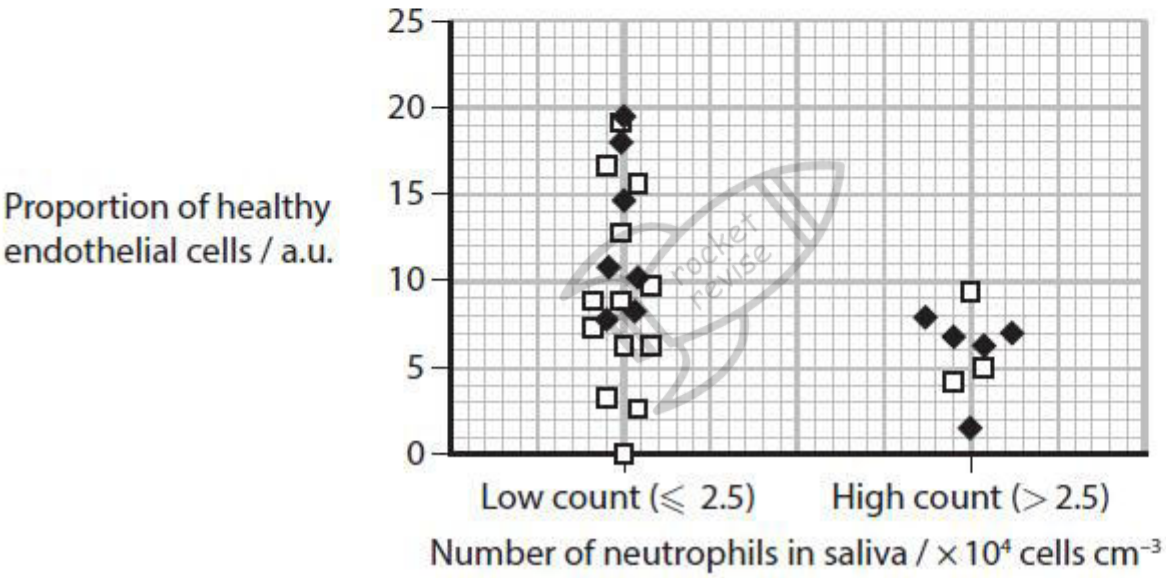
(v) The graphs show some of the results from this study.

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Comment on the relationships between the number of neutrophils and the proportion of healthy endothelial cells of the males and females in this study.

(4)

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(Total for question = 15 marks)

Q24.

Obesity indicators are used to assess the risk of cardiovascular disease (CVD), including coronary heart disease (CHD).

Obesity indicators include body mass index (BMI) and waist-to-hip ratio (WHR).

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(a) Describe the differences between CVD and CHD.

(2)

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(b) A woman who was 165 cm tall had a mass of 68 kg, a waist circumference of 85 cm and a hip measurement of 101 cm.

(i) Calculate the BMI of this woman.
Use the formula:

$$\text{BMI} = \frac{\text{mass in kg}}{(\text{height in m})^2}$$

(1)

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Answer .....

(ii) Calculate the WHR of this woman.

(1)

Answer ..... : 1

(c) Scientists collected data to evaluate these two obesity indicators for their effectiveness in predicting death from CVD and CHD.

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This study involved 9 206 adults aged from 20 to 69 years from Australian cities.
This study started in 1989, when measurements for the obesity indicators were taken.
Risk factors including blood pressure and blood cholesterol levels were measured.
Information about other risk factors was recorded, but not measured.
By the year 2000, 473 of these adults had died. The data were compared with the causes of death.

(i) Explain why high blood pressure is a risk factor for CVD.

(2)

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(ii) Some other risk factors were recorded as being either present or absent but were not measured.
Name **two** other risk factors that might have been recorded but not measured.

(1)

1 .....

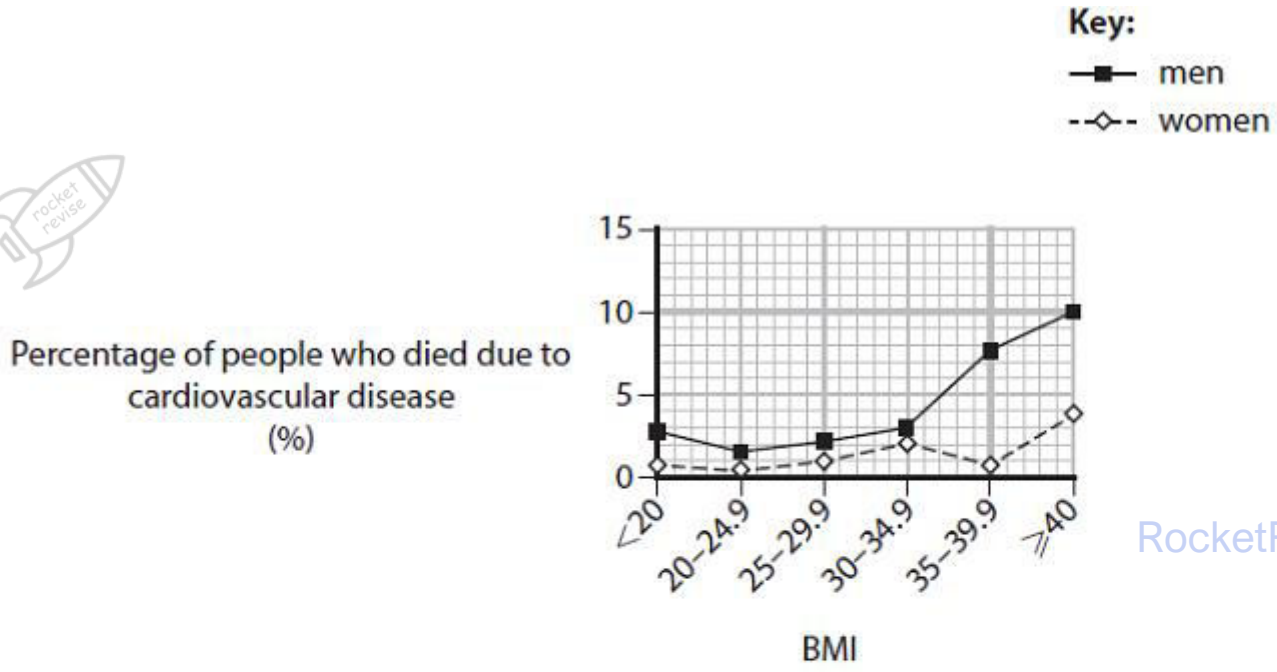
2 .....

(iii) The graphs show the BMI and WHR of the men and women who died due to either CVD or CHD.

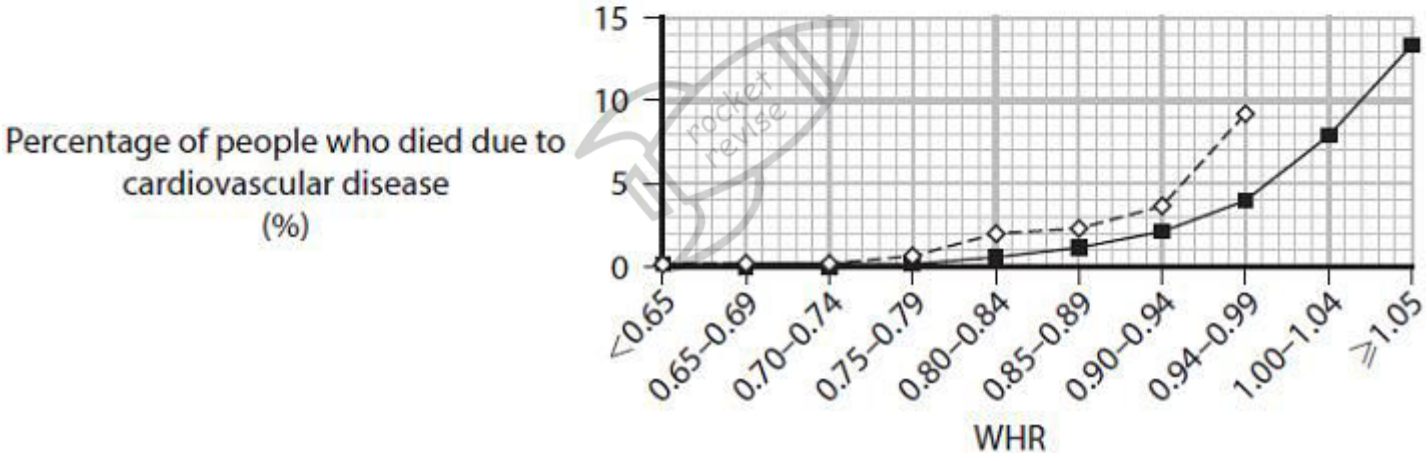
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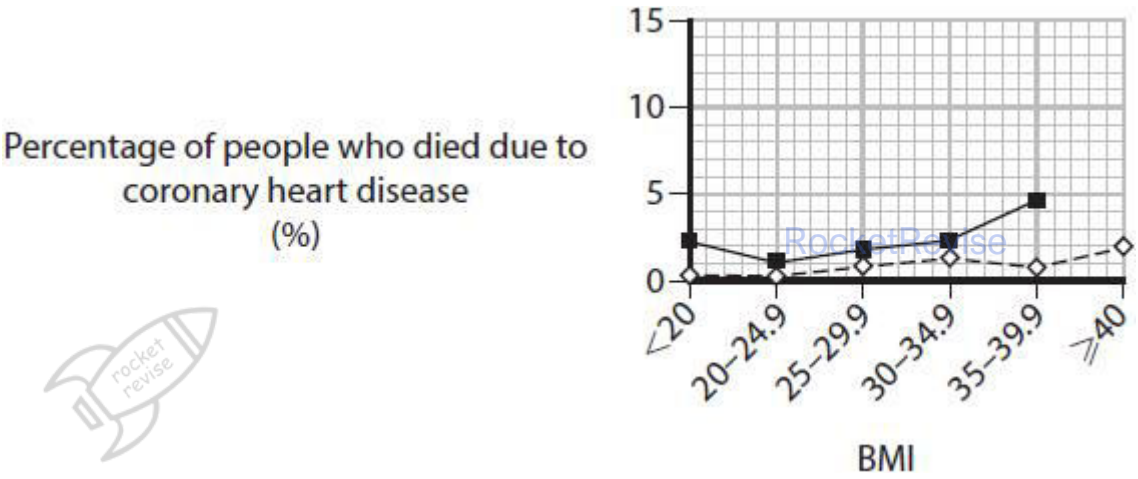
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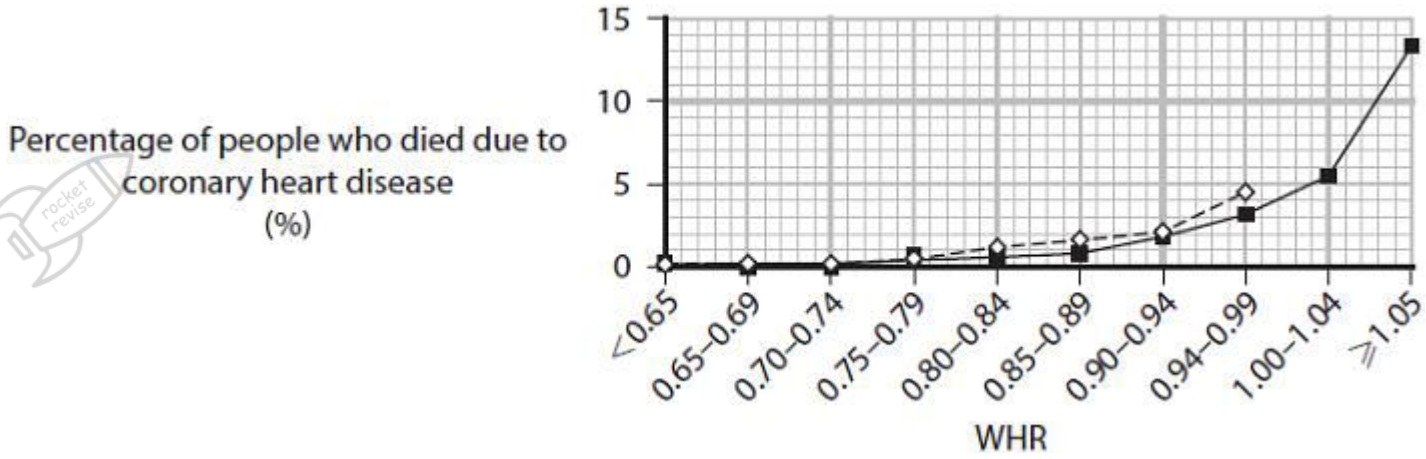
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Discuss the usefulness of these two obesity indicators in predicting death from CVD and CHD.

(3)

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(Total for question = 10 marks)

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